

A nonformal compact manifold with full holonomy $\text{Spin}(7)$ obtained from the Martín–Merchán G_2 construction

Abstract

We start from the flat orbifold underlying the nonformal G_2 manifold of Martín–Merchán and introduce an affine anti- G_2 involution whose translation term is a real quarter-period in the third elliptic coordinate. After adjoining a circle, we couple this involution to reflection in the circle; the resulting involution preserves the product Cayley form. We compute the complete isotropy stratification of the resulting eight-orbifold. In addition to codimension-four A_1 strata, it has codimension-six strata modelled on $\mathbb{R}^2 \times (\mathbb{C}^3/\mathbb{Z}_2^2)$ and isolated strata modelled on $(\mathbb{R}^4/\{\pm 1\})^2$; no isolated $\mathbb{R}^8/\{\pm 1\}$ local model occurs. Explicit Eguchi–Hanson, toric crepant, and product models satisfy all of Joyce’s R -data compatibility conditions and yield a compact torsion-free $\text{Spin}(7)$ resolution. On this resolution we construct a defined triple Massey product of degrees $(3, 2, 2)$ whose full Massey coset does not contain zero, and we prove that the resolution is simply connected. The Berger–Simons–Wang classification then reduces every proper holonomy alternative to a Kähler case, which nonformality excludes. Therefore the resulting metrics have full holonomy $\text{Spin}(7)$.

All cohomology has real coefficients unless otherwise stated. Products of transformations are composed from right to left.

1 Statement of the construction

We set

$$E = \mathbb{C}/(\mathbb{Z} + i\mathbb{Z}), \quad \mathbb{T}^7 = (\mathbb{R}/4\mathbb{Z})_t \times E^3,$$

and write $z_j = x_j + iy_j$. We define

$$F(t, z_1, z_2, z_3) = (t + 1, iz_1, iz_2, -z_3), \tag{1}$$

$$\iota_1(t, z_1, z_2, z_3) = (t, -z_1, -z_2, z_3 + \tfrac{1}{2}), \tag{2}$$

$$\iota_2(t, z_1, z_2, z_3) = (t, -z_1, -z_2, z_3 + \tfrac{i}{2}), \tag{3}$$

$$\kappa(t, z_1, z_2, z_3) = (1 - t, \bar{z}_1, \bar{z}_2, \bar{z}_3). \tag{4}$$

They preserve

$$\varphi = dt \wedge \omega + \text{Re } \Theta, \quad \omega = \sum_{j=1}^3 dx_j \wedge dy_j, \quad \Theta = dz_1 \wedge dz_2 \wedge dz_3. \tag{5}$$

The group $G = \langle F, \iota_1, \iota_2, \kappa \rangle \cong D_4 \times \mathbb{Z}_2^2$ has order 32, and $X = \mathbb{T}^7/G$ is the flat G_2 orbifold of [11].

We define the additional affine involution

$$\tau(t, z_1, z_2, z_3) = (1 - t, z_1, z_2, -z_3 + \tfrac{1}{4}). \tag{6}$$

The translation term $1/4$ in the third complex coordinate is one quarter of the real period of E . We set $S_s^1 = \mathbb{R}/\mathbb{Z}$ and define

$$\Phi_0 = ds \wedge \varphi + *_\varphi \varphi \quad \text{on} \quad \mathbb{T}^8 = S_s^1 \times \mathbb{T}^7. \quad (7)$$

We extend G trivially in the s -direction and define

$$T(s, p) = (-s, \tau p), \quad K = \langle G, T \rangle, \quad \mathcal{O} = \mathbb{T}^8 / K. \quad (8)$$

Theorem 1.1. *There exist a compact smooth eight-manifold $\widehat{Z}$, a proper map $\pi : \widehat{Z} \rightarrow \mathcal{O}$ that is a diffeomorphism over the regular locus, and a number $\varepsilon > 0$ such that, for every $t \in (0, \varepsilon)$, $\widehat{Z}$ carries a torsion-free $\text{Spin}(7)$ structure $\widehat{\Phi}_t$ whose associated metric has full holonomy $\text{Spin}(7)$. Moreover, $\pi_1(\widehat{Z}) = 0$ and $\widehat{Z}$ is nonformal.*

Remark 1.2. The product $S^1 \times M$ of the Martín–Merchán manifold with a circle has a torsion-free $\text{Spin}(7)$ form, but the product metric has holonomy contained in G_2 because its circle vector field is parallel. The construction instead couples the reflection $s \mapsto -s$ with the involution τ , which reverses the G_2 form. Section 5 proves that the resulting resolution is nonformal. Section 6 proves that it is simply connected and then applies the de Rham and Berger–Simons–Wang classifications to show that the holonomy of each sufficiently small torsion-free structure is exactly $\text{Spin}(7)$.

2 The affine group and its fixed sets

2.1 Normalization and the Cayley form

Lemma 2.1. *The map τ normalizes G , with*

$$\tau^2 = 1, \quad \tau F \tau = F^{-1} \iota_1, \quad \tau \iota_j \tau = \iota_j \quad (j = 1, 2), \quad \tau \kappa \tau = \kappa. \quad (9)$$

Moreover,

$$\tau^* \varphi = -\varphi, \quad \tau^* (*_\varphi \varphi) = *_\varphi \varphi. \quad (10)$$

Consequently, $|K| = 64$ and every element of K preserves Φ_0 .

Proof. Substituting the formulas (1)–(4) and (6) into each composite gives all four relations in (9). The derivative of τ sends dt and dz_3 to their negatives and fixes dz_1, dz_2 ; hence it preserves every summand of ω and sends Θ to $-\Theta$. This proves the first equation in (10); the second follows from $*\varphi = \frac{1}{2}\omega^2 - dt \wedge \text{Im } \Theta$. Since $T^*ds = -ds$, one has $T^*\Phi_0 = \Phi_0$. Finally $T \notin G$ and normalizes G , so $|K| = 64$. $\square$

2.2 The coset GT

We set $\bar{t} = t/4$ and use the coordinates

$$q = (\bar{t}, x_1, y_1, x_2, y_2, x_3, y_3) \in (\mathbb{R}/\mathbb{Z})^7. \quad (11)$$

For $q \mapsto Aq + v$, the fixed equation is

$$(A - I)q = -v \pmod{\mathbb{Z}^7}. \quad (12)$$

Lemma 2.2. *Let $M \in M_{m \times n}(\mathbb{Z})$ and $y \in \mathbb{R}^m$. The congruence*

$$Mx = y \pmod{\mathbb{Z}^m}, \quad x \in \mathbb{R}^n/\mathbb{Z}^n,$$

is solvable if and only if

$$\lambda^T y \in \mathbb{Z} \quad \text{for every } \lambda \in \ker(M^T) \cap \mathbb{Z}^m. \quad (13)$$

If it is solvable and the nonzero Smith invariant factors of M are $d_1, \dots, d_r$, then its solution set has dimension $n - r$ and has $d_1 \cdots d_r$ connected components.

Proof. Choose unimodular matrices $U \in \text{GL}(m, \mathbb{Z})$ and $V \in \text{GL}(n, \mathbb{Z})$ such that

$$UMV = \text{diag}(d_1, \dots, d_r, 0, \dots, 0), \quad d_i > 0.$$

The induced coordinate changes are automorphisms of the source and target tori. In these coordinates, the first r equations are $d_i x_i = (Uy)_i$ in $\mathbb{R}/\mathbb{Z}$ and are always solvable, while the remaining equations require $(Uy)_i = 0$ in $\mathbb{R}/\mathbb{Z}$ for $i > r$. These conditions are precisely (13), because the last $m - r$ rows of U form an integral basis of $\ker(M^T) \cap \mathbb{Z}^m$. Each of the first r equations has d_i solutions, and the remaining $n - r$ source coordinates are free. This gives the asserted dimension and number of components. $\square$

Every $g \in G$ is uniquely $F^a \iota_1^b \iota_2^c \kappa^d$. The complete list of $g\tau$ with fixed points is Table 1; signs are in the coordinate order (11). The s -coordinate of gT has sign $-$, so adjoining it neither changes the displayed fixed dimension nor introduces an additional free coordinate.

Table 1: Fixed elements in $G\tau$.

$g\tau$	translation v	sign tuple	dim Fix
τ	$(\frac{1}{4}, 0, 0, 0, 0, \frac{1}{4}, 0)$	$(-, +, +, +, +, -, -)$	4
$\iota_1\tau$	$(\frac{1}{4}, 0, 0, 0, 0, \frac{3}{4}, 0)$	$(-, -, -, -, -, -, -)$	0
$\iota_2\tau$	$(\frac{1}{4}, 0, 0, 0, 0, \frac{1}{4}, \frac{1}{2})$	$(-, -, -, -, -, -, -)$	0
$\kappa\tau$	$(0, 0, 0, 0, 0, \frac{1}{4}, 0)$	$(+, +, -, +, -, -, +)$	4
$\iota_1\kappa\tau$	$(0, 0, 0, 0, 0, \frac{3}{4}, 0)$	$(+, -, +, -, +, -, +)$	4
$\iota_1\iota_2\tau$	$(\frac{1}{4}, 0, 0, 0, 0, \frac{3}{4}, \frac{1}{2})$	$(-, +, +, +, +, -, -)$	4
$F^2\iota_1\iota_2\tau$	$(\frac{3}{4}, 0, 0, 0, 0, \frac{1}{4}, \frac{1}{2})$	$(-, -, -, -, -, -, -)$	0
$F^2\iota_1\tau$	$(\frac{3}{4}, 0, 0, 0, 0, \frac{3}{4}, 0)$	$(-, +, +, +, +, -, -)$	4
$F^2\tau$	$(\frac{3}{4}, 0, 0, 0, 0, \frac{1}{4}, 0)$	$(-, -, -, -, -, -, -)$	0
$F^2\iota_2\tau$	$(\frac{3}{4}, 0, 0, 0, 0, \frac{1}{4}, \frac{1}{2})$	$(-, +, +, +, +, -, -)$	4

Lemma 2.3. *Table 1 is exhaustive. Fourteen nonidentity elements of K have a four-dimensional fixed-point set on $\mathbb{T}^8$. Four have a zero-dimensional fixed-point set. Every remaining nonidentity element acts freely.*

Proof. We write $g = F^a\iota_1^b\iota_2^c\kappa^d$. If $d = 0$ and a is odd, the x_3 equation is translation by an odd quarter-period. If $d = 0$ and a is even, all eight choices have fixed points; four have linear part $-I_7$ and four have a four-dimensional fixed space, exactly as listed. For $d = 1$, solvability of the $\bar{t}$ equation forces $a = 0$, and solvability of the y_3 equation then forces $c = 0$; the only fixed elements in this case are $\kappa\tau$ and $\iota_1\kappa\tau$. Thus six elements of $G\tau$ have four-dimensional fixed sets and four elements of $G\tau$ have zero-dimensional fixed sets. The eight fixed elements of G have three-dimensional fixed sets on $\mathbb{T}^7$, hence four-dimensional fixed sets after adjoining the fixed s -coordinate. Therefore K has fourteen nonidentity elements with four-dimensional fixed sets and four with zero-dimensional fixed sets. $\square$

2.3 The complete isotropy hierarchy

Each of the four elements of $G\tau$ whose fixed set is zero-dimensional has linear part $-I_7$; the corresponding element of GT therefore has linear part $-I_8$ and fixes $2^8 = 256$ points of $\mathbb{T}^8$. These four fixed-point sets are pairwise disjoint. For the four new elements listed in the left column below, the right column lists the unique nonidentity element of G that fixes the same points:

$$\begin{array}{c|c}
 \iota_1 T & \kappa \\
 \iota_2 T & \iota_2 \kappa \\
 F^2 \iota_1 \iota_2 T & F^2 \iota_2 \kappa \\
 F^2 T & F^2 \kappa.
 \end{array} \tag{14}$$

Proposition 2.4 shows that the resulting order-four groups are the full point stabilizers. Every point orbit therefore has $64/4 = 16$ elements, and the $4 \cdot 256$ fixed points determine $1024/16 = 64$ isolated points in $\mathbb{T}^8/K$.

Four subgroups have two-dimensional common fixed sets:

$$A_0 = \langle \kappa, T \rangle, \quad A_1 = \langle \iota_2 \kappa, \iota_1 \iota_2 T \rangle, \tag{15}$$

$$A_2 = \langle \kappa T, F^2 \iota_2 \kappa \rangle, \quad A_3 = \langle F^2 \kappa, \iota_1 \kappa T \rangle. \tag{16}$$

Each common fixed set is a disjoint union of 64 two-tori. The subgroups form two K -conjugacy classes of size two.

Proposition 2.4. *For every $p \in \mathbb{T}^8$ with nontrivial stabilizer K_p , exactly one of the following three alternatives holds. First, $K_p \cong \mathbb{Z}_2$ and acts as $(I_4, -I_4)$. Second, p lies on a two-dimensional stratum, $K_p \cong \mathbb{Z}_2^2$, and the three nonidentity elements act on the six-dimensional normal space by the nontrivial even sign changes; the local model is*

$$\mathbb{R}^2 \times \mathbb{C}^3 / \mathbb{Z}_2^2. \quad (17)$$

Third, p is one of the 64 isolated quotient points, $K_p \cong \mathbb{Z}_2^2$, and the local model is

$$(\mathbb{R}^4 / \{\pm 1\}) \times (\mathbb{R}^4 / \{\pm 1\}). \quad (18)$$

There are no larger stabilizers.

Proof. For affine elements $h_\nu(q) = A_\nu q + v_\nu$, a common fixed point is a solution of the combined congruence

$$\begin{pmatrix} A_1 - I \\ \vdots \\ A_r - I \end{pmatrix} q = - \begin{pmatrix} v_1 \\ \vdots \\ v_r \end{pmatrix} \pmod{\mathbb{Z}^{8r}}. \quad (19)$$

For each intersecting pair (h_1, h_2) , we record the fixed-dimension data

$$(\dim \text{Fix } h_1, \dim \text{Fix } h_2, \dim \text{Fix}(h_1 h_2); \dim(\text{Fix } h_1 \cap \text{Fix } h_2)).$$

Lemma 2.2 applies to the block matrix in (19). Its criterion and Smith normal form give exactly twelve pairs with data $(4, 4, 4; 2)$. These pairs assemble three at a time into the four subgroups (15)–(16). For each subgroup, the Smith normal form has component group of order 64, so its common fixed set lifts to 64 two-tori. Every other nonempty pairwise intersection belongs to one of the four order-four point stabilizers in (14). Repeating the calculation for triples yields no additional system: there are exactly four systems of type $(4, 4, 4)$ and four of type $(4, 4, 0)$. Testing the remaining elements of K on these solution sets shows that none fixes a point of any set, so no stabilizer is larger.

At a two-dimensional stratum, the normal representation is the direct sum of the three nontrivial real characters of $\mathbb{Z}_2^2$, each with multiplicity two. After choosing a complex coordinate on each character plane, the three nonidentity elements act by the even sign changes in (17). At an isolated point, we denote the unique nonidentity stabilizer element in G by a and the element of GT with zero-dimensional fixed set by b . The eigenspace decomposition of a gives coordinates in which

$$a = (I_4, -I_4), \quad b = (-I_4, -I_4), \quad ab = (-I_4, I_4),$$

so quotienting by $\langle a, b \rangle$ gives the product model (18). $\square$

3 Joyce resolution data

We use Joyce's notation [10, Definition 13.2.2]. For a connected fixed component C , we define

$$N_K(C) = \{\gamma \in K : \gamma C = C\}, \quad C_K(C) = \{\gamma \in K : \gamma|_C = \text{id}_C\}.$$

We denote the generic exact isotropy along C by A_C , its tangent space by V_C , and its orthogonal complement by $W_C = V_C^\perp$. Because C is a connected component of the fixed set of A_C and its generic points have no larger stabilizer, $C_K(C) = A_C$. Joyce's residual component group is therefore

$$B_C = N_K(C) / C_K(C) = N_K(C) / A_C. \quad (20)$$

This is different from the subgroup normalizer $N_K(A_C)$: in general $N_K(C) \subset N_K(A_C)$. If $C = \{p\}$ is a point, then $N_K(p) = C_K(p)$ and $B_p = 1$.

3.1 The four-dimensional strata

We use the coframe

$$(e^0, e^1, e^2, e^3, e^4, e^5, e^6, e^7) = (ds, dt, dx_1, dy_1, dx_2, dy_2, dx_3, dy_3). \quad (21)$$

We write $(e_0, \dots, e_7)$ for its dual frame. For an oriented normal basis $(w_1, \dots, w_4)$, we define $\sigma_1 = w^{12} + w^{34}$, $\sigma_2 = w^{13} - w^{24}$, $\sigma_3 = w^{14} + w^{23}$. We also define $p_{\pm} = (e_2 \pm e_3)/\sqrt{2}$, $q_{\pm} = (e_4 \pm e_5)/\sqrt{2}$.

Table 2: Normal data for the six four-stratum conjugacy classes.

h	$ h $	$ Z_K(h) $	V_h	W_h and ζ_h
κ	2	32	(e_0, e_2, e_4, e_6)	$(e_1, e_7, e_3, e_5); e^{17} + e^{35}$
T	2	32	(e_2, e_3, e_4, e_5)	$(e_0, e_6, e_1, -e_7); e^{06} - e^{17}$
κF	4	16	(e_0, p_-, q_-, e_7)	$(e_1, -e_6, p_+, q_+);$ $-e^{16} + \frac{1}{2}(e^2 + e^3) \wedge (e^4 + e^5)$
$\iota_2 \kappa$	2	32	$(e_0, e_3, e_5, -e_6)$	$(e_1, -e_7, e_2, e_4); -e^{17} + e^{24}$
κT	2	32	$(e_1, e_2, e_4, -e_7)$	$(e_0, -e_6, e_3, e_5); -e^{06} + e^{35}$
$\iota_1 \iota_2 T$	2	32	(e_2, e_3, e_4, e_5)	$(e_0, e_6, e_1, -e_7); e^{06} - e^{17}$

The rows of Table 2 classify conjugacy classes of generators, rather than connected fixed components. We denote by $\mathcal{F}_4$ the finite set of connected components with generic exact isotropy $\mathbb{Z}_2$. For each K -orbit in $\mathcal{F}_4$, we choose one component. Its generator is conjugate to exactly one row of the table, and the displayed linear data are attached to that component and then transported around its orbit. We make the same orbitwise choice for the components of the four two-dimensional common fixed sets and for the isolated points. Transporting these choices assigns resolution data to every connected component of every stratum.

Lemma 3.1. *Each V_h is Cayley-oriented, ζ_h is self-dual on W_h , and the image of $Z_K(h)$ on $\Lambda^+ W_h^*$ is*

$$\{I_3, \text{diag}(1, -1, -1)\} \quad (22)$$

in a basis beginning with ζ_h .

Proof. The expanded Cayley form is

$$\begin{aligned} \Phi_0 = & e^{0123} + e^{0145} + e^{0167} + e^{0246} - e^{0257} - e^{0347} - e^{0356} \\ & + e^{2345} + e^{2367} + e^{4567} - e^{1247} - e^{1256} - e^{1346} + e^{1357}. \end{aligned} \quad (23)$$

For each row of Table 2, evaluation of the explicit form (23) on the listed oriented basis of V_h is $+1$; hence that orientation is the Cayley orientation. Relative to the listed oriented basis of W_h , the form ζ_h equals $w^{12} + w^{34}$, so it is self-dual and is the first vector of the standard oriented basis of $\Lambda^+ W_h^*$. We restrict the five generator matrices to W_h and write a general element in the form $q = F^a \iota_1^b \iota_2^c \kappa^d T^e$. For $h = \kappa, T, \iota_2 \kappa, \kappa T, \iota_1 \iota_2 T$, the equation $qh = hq$ is exactly $a \equiv 0 \pmod{2}$, giving 32 elements. For $h = \kappa F$ it is exactly $e = 0$ and $d \equiv a \pmod{2}$, giving 16 elements. Thus the centralizer orders are 32 and 16, respectively, and orbit-stabilizer gives the class sizes displayed in Table 2. The induced action of each centralizer on $(\sigma_1, \sigma_2, \sigma_3)$ has image $\{I_3, \text{diag}(1, -1, -1)\}$. The nontrivial image is realized by κ for $h = T$, by ι_1 for $h = \kappa F$, and by the corresponding directly computed centralizer element in each remaining row. $\square$

We fix numbers $L > 2\ell > 0$. For a four-dimensional component C with pointwise stabilizer generated by h_C , we define

$$s(C) = \begin{cases} L, & h_C \in G, \\ \ell, & h_C \in GT. \end{cases} \quad (24)$$

We equip the radial Eguchi–Hanson resolution $Y_C = \text{EH}_{s(C)} \rightarrow W_C/\{\pm 1\}$ with the normalized Kähler form whose ALE limit lies on the ray $\mathbb{R}_{>0}\zeta_C$ and whose integral over the exceptional sphere is $s(C)$. To specify the full product $\text{Spin}(7)$ structure, we must also fix the remaining two members of its hyperkähler triple. We write $(v_1, \dots, v_4)$ for the ordered frame of V_C in Table 2, and we define

$$\alpha_1 = v^{12} + v^{34}, \quad \alpha_2 = v^{13} - v^{24}, \quad \alpha_3 = v^{14} + v^{23}.$$

We choose the oriented self-dual basis $(\sigma_1, \sigma_2, \sigma_3)$ of W_C^* beginning with $\sigma_1 = \zeta_C$ for which the flat splitting of (23) is

$$\Phi_0 = v^{1234} + \alpha_1 \wedge \sigma_1 + \alpha_2 \wedge \sigma_2 - \alpha_3 \wedge \sigma_3 + \frac{1}{2}\sigma_1^2. \quad (25)$$

We denote by $(\omega_{C,1}, \omega_{C,2}, \omega_{C,3})$ the parallel hyperkähler triple on $\text{EH}_{s(C)}$ that is asymptotic to $(\sigma_1, \sigma_2, \sigma_3)$ and whose class $[\omega_{C,1}]$ evaluates to $s(C)$ on the exceptional sphere. The complete product model used below is therefore

$$\Phi_C = v^{1234} + \alpha_1 \wedge \omega_{C,1} + \alpha_2 \wedge \omega_{C,2} - \alpha_3 \wedge \omega_{C,3} + \frac{1}{2}\omega_{C,1}^2, \quad g_C = g_{V_C} + g_{\text{EH}_{s(C)}}. \quad (26)$$

Equation (26) is Joyce’s product formula from [10, Proposition 13.1.1] and specifies the complete metric and Cayley form at the four-dimensional stratum. Because $G \triangleleft K$, conjugation preserves the alternatives in (24). Thus $s(C) = s(C')$ whenever $\gamma C = C'$. If $\gamma C = C'$, we denote by $\rho_+(D\gamma|_{W_C})$ the transport on self-dual two-forms induced by the oriented isometry $D\gamma : W_C \rightarrow W_{C'}$, and we define

$$\zeta_{C'} = \epsilon_{\gamma,C} \rho_+(D\gamma|_{W_C}) \zeta_C, \quad \epsilon_{\gamma,C} \in \{\pm 1\}. \quad (27)$$

If $\epsilon_{\gamma,C} = +1$, we lift $D\gamma|_{W_C}$ by the radial holomorphic Eguchi–Hanson isometry. If $\epsilon_{\gamma,C} = -1$, we use the radial antiholomorphic lift. Definition (27) gives the cocycle identity

$$\epsilon_{\gamma\delta,C} = \epsilon_{\gamma,\delta C} \epsilon_{\delta,C}. \quad (28)$$

Lemma 3.1 implies $\epsilon_{\gamma,C} = 1$ for every $\gamma \in N_K(C)$, so transport from an orbit representative is independent of the transporting element. Each of the two possible lifts preserves the Eguchi–Hanson metric and induces on its hyperkähler triple the same orthogonal transformation that $D\gamma$ induces on $\Lambda^+ W_C^*$. Because $D\gamma$ preserves the flat identity (25), its action on $(\alpha_1, \alpha_2, \alpha_3)$ is paired with the normal action so that every term in (26) is fixed. Therefore the transported four-stratum metric and Cayley form are exactly K -equivariant.

3.2 The two- and zero-dimensional strata

For $A_0 = \langle \kappa, T \rangle$, $V_{A_0} = \langle e_2, e_4 \rangle$ and a unitary normal coframe is

$$\xi_1 = e^0 + ie^6, \quad \xi_2 = e^3 - ie^5, \quad \xi_3 = e^1 - ie^7. \quad (29)$$

Relative to (ξ_1, ξ_2, ξ_3) , the elements κ and T act by the sign triples $(1, -1, -1)$ and $(-1, 1, -1)$, respectively. For A_1 , whose common plane is $\langle e_3, e_5 \rangle$, we use the unitary normal coframe

$$\eta_1 = e^0 - ie^6, \quad \eta_2 = e^2 - ie^4, \quad \eta_3 = e^1 + ie^7. \quad (30)$$

The subgroups A_2 and A_3 are K -conjugate to A_1 and A_0 , respectively. Hence every two-stratum normal representation is conjugate in $SU(3)$ to the diagonal subgroup

$$A = \{(1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1)\} \subset SU(3). \quad (31)$$

The corresponding flat Calabi–Yau forms are

$$\omega_{A_0} = e^{06} - e^{35} - e^{17}, \quad \Omega_{A_0} = \xi_1 \wedge \xi_2 \wedge \xi_3, \quad (32)$$

$$\omega_{A_1} = -e^{06} - e^{24} + e^{17}, \quad \Omega_{A_1} = \eta_1 \wedge \eta_2 \wedge \eta_3. \quad (33)$$

We choose the oriented tangent coframe $(u^1, u^2) = (-e^2, -e^4)$ for the A_0 -stratum and $(u^1, u^2) = (e^3, e^5)$ for the A_1 -stratum. Expanding the right-hand side of the following identity with (32)–(33) reproduces every term of (23); therefore, for $r = 0, 1$,

$$\Phi_0 = u^{12} \wedge \omega_{A_r} + \operatorname{Re}((u^1 + iu^2) \wedge \Omega_{A_r}) + \frac{1}{2}\omega_{A_r}^2. \quad (34)$$

Suppose that a component fixed by A_r lies in a four-dimensional component whose generic stabilizer is generated by $h \in A_r$. We denote the complex normal coordinate fixed by h by λ_h , and we denote the direct sum of the other two complex coordinate lines by Q_h . If Z_h is the unique coordinate $(1, 0)$ -vector satisfying $d\lambda_h(Z_h) = 1$, we define the complete ordered flat transverse triple by

$$\varpi_{h,1} = \omega_{A_r}|_{Q_h}, \quad \varpi_{h,2} + i\varpi_{h,3} = (\iota_{Z_h}\Omega_{A_r})|_{Q_h}. \quad (35)$$

Equivalently, $\varpi_{h,1}$ is the restriction of ω_{A_r} to Q_h , while $\varpi_{h,2}$ and $\varpi_{h,3}$ are the real and imaginary parts of the two-form obtained by contracting Ω_{A_r} with Z_h . With $\zeta_{\iota_1\kappa T} = -e^{06} - e^{24}$ (using the oriented normal basis $(e_0, -e_6, e_2, -e_4)$), the six restrictions are

	h : fixed complex line, $\omega_{\perp,h}$		
A_0	$\kappa : \xi_1, -\zeta_\kappa$	$T : \xi_2, +\zeta_T$	$\kappa T : \xi_3, -\zeta_{\kappa T}$
A_1	$\iota_2\kappa : \eta_1, -\zeta_{\iota_2\kappa}$	$\iota_1\iota_2T : \eta_2, -\zeta_{\iota_1\iota_2T}$	$\iota_1\kappa T : \eta_3, +\zeta_{\iota_1\kappa T}$

(36)

In each row the first involution belongs to G and its transverse Eguchi–Hanson factor has scale L ; the other two belong to GT and have scale ℓ . For every incidence in (36), the line $\mathbb{R}\varpi_{h,1}$ equals the distinguished line $\mathbb{R}\zeta_h$ of the incident four-stratum, and the table records the orientation sign. When the sign is positive, we identify the transverse resolution with the incident Eguchi–Hanson space by the radial holomorphic lift. When the sign is negative, we use the radial antiholomorphic lift and choose the phase of its holomorphic symplectic form so that the pullback of $(\omega_{C,1}, \omega_{C,2}, \omega_{C,3})$ equals $(\varpi_{h,1}, \varpi_{h,2}, \varpi_{h,3})$. Both triples are parallel and have the same ALE limit under this identification, so their difference is a parallel tensor that tends to zero and therefore vanishes identically. The incidences for $A_3 = FA_0F^{-1}$ and $A_2 = FA_1F^{-1}$ are defined by this F -transport, and their orientation signs are determined by (28).

In each of the coframes (29) and (30), the unique nonidentity element of $A_r \cap G$ acts by $(1, -1, -1)$ and fixes the first complex coordinate line. We define $m_{ij} = (e_i + e_j)/2$ and

$$N_a = \mathbb{Z}^3 + \mathbb{Z}m_{23}, \quad N = N_a + \mathbb{Z}m_{13} = \mathbb{Z}^3 + \mathbb{Z}m_{12} + \mathbb{Z}m_{13}. \quad (37)$$

We first quotient by the old subgroup $A_r \cap G = \langle(1, -1, -1)\rangle$. Its crepant subdivision is taken in N_a and has the two maximal cones

$$\langle e_1, e_2, m_{23} \rangle, \quad \langle e_1, m_{23}, e_3 \rangle.$$

Passing to the residual involution replaces N_a by the quotient lattice N . The residual fixed curves correspond to the two faces $\langle e_1, e_2 \rangle$ and $\langle e_1, e_3 \rangle$; star subdivision of these faces introduces the primitive rays m_{12} and m_{13} and gives the staged fan

$$\Sigma_{\text{st}}^{\max} = \{ \langle e_1, m_{12}, m_{23} \rangle, \langle e_2, m_{12}, m_{23} \rangle, \langle e_1, m_{13}, m_{23} \rangle, \langle e_3, m_{13}, m_{23} \rangle \}. \quad (38)$$

For each of the four maximal cones, the determinant of its primitive ray generators in the standard lattice is $\pm 1/4$, which equals the covolume of N . Those generators therefore form a basis of N , and the fan is smooth. Every primitive ray generator lies on the height-one hyperplane, so the subdivision is crepant. Because the subdivision has the same support $\mathbb{R}_{\geq 0}^3$ as the affine quotient fan, the associated toric morphism is proper and birational. Hence

$$Y_A = Y_{\Sigma_{\text{st}}} \longrightarrow \mathbb{C}^3/A \quad (39)$$

is smooth and crepant. By [10, Proposition 9.1.4], every crepant resolution of $\mathbb{C}^m/H$ is a local-product resolution in the sense required by Joyce's R -data definition.

We denote generic exceptional $\mathbb{CP}^1$ fibres transverse to the first, second, and third coordinate axes by C_{23}, C_{13}, C_{12} , respectively. Consider the torus-invariant real divisor

$$D = -\frac{L}{2}D_{m_{23}} - \frac{\ell}{2}(D_{m_{12}} + D_{m_{13}}).$$

In Fulton's convention, the support function of $D = \sum_{\rho} a_{\rho} D_{\rho}$ satisfies $\psi_D(v_{\rho}) = -a_{\rho}$. Thus the support function associated with D is defined on the primitive rays of Σ_{st} by

$$\psi_D(e_i) = 0, \quad \psi_D(m_{23}) = \frac{L}{2}, \quad \psi_D(m_{12}) = \psi_D(m_{13}) = \frac{\ell}{2}. \quad (40)$$

The compact walls $\langle m_{12}, m_{23} \rangle$, $\langle m_{13}, m_{23} \rangle$, and $\langle e_1, m_{23} \rangle$ are exactly the internal two-cones of the fan and have wall degrees ℓ , ℓ , and $(L - 2\ell)/2$, respectively. For the third wall, this computation uses $m_{12} + m_{13} = e_1 + m_{23}$. Since $L > 2\ell > 0$, all three wall degrees are positive, and ψ_D is strictly upper convex in Fulton's convention. Restriction to the three transverse A_1 fans, whose primitive relations are $e_i + e_j = 2m_{ij}$, gives the QALE Kähler class periods

$$\int_{C_{23}} \omega_A = L, \quad \int_{C_{13}} \omega_A = \int_{C_{12}} \omega_A = \ell. \quad (41)$$

We normalize the compact torus as $N_{\mathbb{R}}/N$, so every primitive angular coordinate has period one. In a convention in which primitive angles have period 2π , the support values must instead be divided by 2π ; with either convention the geometric periods are those in (41). The toric support-function theorem [7, §3.4] says that strict upper convexity is the toric ampleness criterion. In the present noncompact situation this is first a relative statement for $Y_A \rightarrow \mathbb{C}^3/A$, because the fan has the support of the single affine cone. Since the base is affine, a relatively ample rational divisor is ample on Y_A ; openness of the ample cone and linearity extend this conclusion to the real divisor D . Its real first Chern class is therefore a Kähler class, and its integrals over invariant compact curves are the wall pairings above. The pair $(A, \mathbb{C}^3/A)$ is the one treated in [10, Example 9.3.6], which proves that each of its four crepant resolutions admits a QALE Kähler metric. The toric variety Y_A is one of these resolutions and is local-product by Proposition 9.1.4. Joyce's [10, Theorem 9.4.7] therefore applies to the specific Kähler class defined by D , equivalently by (40), and supplies a QALE Kähler representative of that class.

We apply Joyce's Theorem 9.3.3 to the Kähler class just constructed and denote its unique Ricci-flat QALE representative by g_A , with Kähler form ω_A . We normalize the parallel holomorphic volume form Ω_A so that its limit is (32) or (33), according to the two-stratum orbit. The resulting Calabi–Yau threefold $(Y_A, g_A, \omega_A, \Omega_A)$ has one singular-axis end asymptotic to $\mathbb{C} \times \text{EH}_L$ and two ends asymptotic to $\mathbb{C} \times \text{EH}_\ell$, with the hyperkähler identifications in (36).

The existence of Ω_A follows from the crepant toric construction. The invariant flat volume form on the regular locus of $\mathbb{C}^3/A$ pulls back to the inverse image of that locus in Y_A . The height-one condition implies that its divisor along every toric exceptional divisor is zero, so it extends to a nowhere-vanishing holomorphic three-form on all of Y_A . Joyce's [10, Example 9.3.6 and Theorem 9.3.4] identifies the holonomy of g_A with $SU(3)$; hence the extended volume form is parallel and unique up to a nonzero constant, whose modulus and phase are fixed by the prescribed flat-end limit. Proposition 13.1.2 and Theorem 13.1.10(a) of [10] associate to these Calabi–Yau data the required torsion-free product $\text{Spin}(7)$ structure on $\mathbb{R}^2 \times Y_A$.

At an isolated stratum, we denote the nonidentity old stabilizer by $a \in G$ and the all-minus stabilizer element by $b \in GT$. We choose the eigenspace splitting $\mathbb{R}^8 = U \oplus U'$ for which $a = (I_U, -I_{U'})$, $b = (-I_U, -I_{U'})$, and $ab = (-I_U, I_{U'})$, and we use the product resolution

$$\text{EH}_\ell(U) \times \text{EH}_L(U') \longrightarrow (U/\{\pm 1\}) \times (U'/\{\pm 1\}). \quad (42)$$

We choose quaternionic structures on U and U' that are compatible with Φ_0 . Under these identifications, the stabilizer is the sign subgroup of $Sp(1) \times Sp(1) \subset Sp(2)$. We choose parallel triples $(\omega_1, \omega_2, \omega_3)$ and $(\omega'_1, \omega'_2, \omega'_3)$ on the two Eguchi–Hanson factors and normalize their flat limits so that the following expression tends to (23):

$$\begin{aligned} \Phi_p &= \frac{1}{2}(\omega_1 + \omega'_1)^2 + \text{Re}((\omega_2 + i\omega_3) \wedge (\omega'_2 + i\omega'_3)) \\ &= \text{vol}_{\text{EH}_\ell} + \text{vol}_{\text{EH}_L} + \omega_1 \wedge \omega'_1 + \omega_2 \wedge \omega'_2 - \omega_3 \wedge \omega'_3. \end{aligned} \quad (43)$$

The Riemannian metric associated with Φ_p is the product metric

$$g_p = g_{\text{EH}_\ell(U)} + g_{\text{EH}_L(U')}. \quad (44)$$

Equation (43) is Proposition 13.1.4 applied to the product Calabi–Yau four-fold, whose Kähler form is $\omega_1 + \omega'_1$ and whose holomorphic volume form is $(\omega_2 + i\omega_3) \wedge (\omega'_2 + i\omega'_3)$. Hence Φ_p is closed and torsion-free. It remains to verify the QALE estimates for the unequal-scale product. Write $r_U = 1 + |u|$ and $r_{U'} = 1 + |u'|$. On the end modelled on $U \times \text{EH}_L(U')$, the ALE chart replaces the $\text{EH}_\ell(U)$ factor by its flat cone and gives, for every $k \geq 0$,

$$\nabla^k(\Phi_p - \Phi_{U \times \text{EH}_L}) = O(r_U^{-4-k});$$

on the end modelled on $\text{EH}_\ell(U) \times U'$, one similarly has

$$\nabla^k(\Phi_p - \Phi_{\text{EH}_\ell \times U'}) = O(r_{U'}^{-4-k}).$$

We compare these estimates with the weights in [10, Definition 13.1.8]. In the first chart the index i has fixed space $V_i = U$. Let j be the index of the complementary fixed space $V_j = U'$. Then $W_j = U$ has real dimension four, so Joyce's exponent is $d_j = 2 - \dim_{\mathbb{R}} W_j = -2$. The subset V_j/A_i of $V_i \times W_i/A_i$ is $\{0\} \times U'/\{\pm 1\}$; hence $\mu_{i,j} \asymp r_U$ on the QALE region. The minimum defining ν_i also contains $\mu_{i,i}$, which measures the transverse radius and can be smaller than r_U . What is needed is the one-sided estimate $1 \leq \nu_i \leq 1 + \mu_{i,j} \leq Cr_U$. Since the exponents are negative, this gives

$$r_U^{-4-k} \leq C\mu_{i,j}^{d_j}\nu_i^{-2-k}.$$

Thus the first ALE error is O of the weight required by Joyce's definition, even when the required weight is larger than r_U^{-4-k} . The second chart is identical after interchanging U and U' .

In the ambient chart, the two four-plane indices satisfy

$$\mu_{\mathbf{0},U} \asymp r_{U'}, \quad \mu_{\mathbf{0},U'} \asymp r_U, \quad \nu_{\mathbf{0}} \asymp 1 + \min(|u|, |u'|),$$

and both exponents are -2 . If, for example, $r_U \geq r_{U'}$, then every derivative of the error contributed by the U -factor, every derivative of the error contributed by the U' -factor, and every mixed product of the two errors is bounded by a constant multiple of $r_{U'}^{-4-k}$. This is the summand $O(\mu_{\mathbf{0},U}^{-2} \nu_{\mathbf{0}}^{-2-k})$; the case $r_{U'} \geq r_U$ is symmetric. Thus, for every $k \geq 0$, the full error is bounded by

$$\sum_{q: \mathbf{0} \not\leq q} O(\mu_{\mathbf{0},q}^{d_q} \nu_{\mathbf{0}}^{-2-k}).$$

The remaining, deeper-stratum summands on the right are harmless because the asserted estimate is an upper bound by the whole sum. This verifies Definition 13.1.8 in every chart for $\text{EH}_\ell \times \text{EH}_L$.

Proposition 3.2. *The four-stratum Eguchi–Hanson models, the two-stratum toric models, the isolated-point product models, and the transition and transport maps specified above and in the proof below constitute R -data for $\mathbb{T}^8/K$ in the sense of [10, Definition 13.2.2].*

Proof. We verify the three compatibility requirements of Joyce's definition. Condition (i) is verified separately in each stratum dimension. At a four-stratum we use the product $\mathbb{R}^4 \times \text{EH}_{s(C)}$ of [10, Proposition 13.1.1]. At a two-stratum we use $\mathbb{R}^2 \times Y_A$ and [10, Proposition 13.1.2]. At an isolated point we use the product (42) and [10, Proposition 13.1.4]. The estimates following (42) prove directly that the isolated-point model is QALE, while [10, Theorem 13.1.10(a)] proves the QALE assertion for $\mathbb{R}^2 \times Y_A$.

For condition (ii), the face $\langle e_2, e_3 \rangle$ transverse to the distinguished first axis is subdivided by m_{23} and has period L ; the other two coordinate faces are subdivided by m_{13}, m_{12} and have period ℓ . Each transverse fan is the minimal A_1 fan, so the three models are respectively $\text{EH}_L, \text{EH}_\ell, \text{EH}_\ell$. By [10, Proposition 9.1.4], Y_A is a local-product resolution. For each fixed coordinate axis, we choose the map required by [10, Definition 9.1.2] to be the toric product map obtained from the star of the corresponding face,

$$\psi_{A,h} : \mathcal{U}_{A,h} \longrightarrow Y_A. \quad (45)$$

Here h ranges over the three nonidentity elements of A . We set $s_h = L$ when $h \in A \cap G$, and we set $s_h = \ell$ for the other two elements of A . The set $\mathcal{U}_{A,h}$ is the complement of the deeper-stratum set in the product of the coordinate axis with the transverse toric A_1 -resolution. The star construction identifies it with an open neighborhood of the corresponding end of Y_A , and $\psi_{A,h}$ commutes with the blowdowns. Proposition 9.1.5 says that the induced transverse resolution is itself local-product. We use these maps, with the holomorphic or antiholomorphic transverse identification specified after (36). Formula (35) then identifies the entire ordered hyperkähler triple, and substitution into (26) and (49) identifies the complete Cayley forms. At an isolated point, the transition to the four-stratum modelled on $U \times \text{EH}_L(U')$ is the product of the inverse ALE chart

$$(U/\{\pm 1\}) \setminus B_R \longrightarrow \text{EH}_\ell(U)$$

with the identity on $\text{EH}_L(U')$. The transition to the end modelled on $\text{EH}_\ell(U) \times U'$ uses the inverse ALE chart $(U'/\{\pm 1\}) \setminus B_R \rightarrow \text{EH}_L(U')$ in the second factor and the identity in the first factor. The normal factors have scales ℓ and L , matching the new and old incident four-strata, respectively.

We next identify the lower-dimensional metrics occurring in the QALE asymptotics. In a QALE chart along one of the three singular axes, Definition 9.2.1 supplies a lower-dimensional QALE Kähler metric $g_{A,h}$ on the transverse minimal A_1 resolution; Proposition 9.2.4 says that this asymptotic metric is itself QALE. Because g_A is Ricci-flat and the QALE convergence holds with two derivatives, the Ricci tensors in the rescaled charts converge to $\text{Ric}(g_{A,h})$; hence $g_{A,h}$ is Ricci-flat. Restricting (41) to the transverse exceptional sphere gives area L or ℓ . As $H^2(\text{EH}; \mathbb{R}) = \mathbb{R} \text{PD}[E]$, where E is the exceptional sphere, this single period determines the complete Kähler class, and the uniqueness part of Theorem 9.3.3 gives

$$g_{A,h} = g_{\text{EH}_L} \quad \text{or} \quad g_{A,h} = g_{\text{EH}_\ell}$$

with the ALE normalization already chosen at the incident four-stratum. Moreover, the normalized crepant volume form in the local-product chart is the wedge of the coordinate-axis one-form with the Eguchi–Hanson holomorphic two-form. Substitution of this splitting into Joyce’s formula in Proposition 13.1.2 proves the complete all-derivative $\text{Spin}(7)$ asymptotic required by Definition 13.1.8. For the isolated-point model, the same all-derivative conclusion follows from the explicit product estimates above.

In Joyce’s partial order, we denote the ambient nonsingular index by $\mathbf{0}$. Every singular index satisfies $i \succ \mathbf{0}$. The remaining relations are exactly the following: each two-stratum index lies above the three incident four-stratum indices in (36), and each isolated-point index lies above the two four-stratum indices associated with U and U' . If a chain contains a two-stratum, the corresponding transition map is the composition through the appropriate axis chart (45). Proposition 2.4 shows that no other stabilizer, and hence no other comparable pair or chain, occurs.

For condition (iii), we must prove exact equivariance of both the metric and the Cayley form. Lemma 3.1 together with (27) establishes this equivariance at every four-stratum. For the representatives A_0, A_1 , direct multiplication in the normal form $F^a \iota_1^b \iota_2^c \kappa^d T^e$ shows that membership in either subgroup normalizer is equivalent to $a \equiv 0 \pmod{2}$, which gives

$$N_K(A_0) = N_K(A_1) = \{F^a \iota_1^b \iota_2^c \kappa^d T^e : a \equiv 0 \pmod{2}\}. \quad (46)$$

Thus each subgroup normalizer has order 32 and is generated by $F^2, \iota_1, \iota_2, \kappa, T$. Their complex signs in the coframes (29)–(30) are

	(ξ_1, ξ_2, ξ_3)	(η_1, η_2, η_3)	
F^2, ι_1, ι_2	+ − +	+ − +	
κ	+ − −	+ + −	
T	− + −	− + −	

(47)

The derivative of each listed generator is diagonal and complex-linear in the indicated coframe. It therefore acts trivially on the fan lattice, fixes each of the six ray labels in (38), and preserves every support value in (40). Since the normalizer of a connected component is a subgroup of the subgroup normalizer, these properties also hold for every element of its component group $B_{C_A} = N_K(C_A)/A$. A diagonal complex sign map is an automorphism of the dense algebraic torus and fixes every cone of Σ_{st} ; it therefore has a unique holomorphic toric lift $\hat{\gamma}$ to Y_A agreeing with it on that dense torus.

The toric lift $\hat{\gamma}$ preserves the Kähler class specified by (41). Hence $\hat{\gamma}^* \omega_A$ is a Ricci-flat QALE Kähler form in the same class and with the same normalized flat limit as ω_A ; uniqueness in Joyce’s Theorem 9.3.3 gives

$$\hat{\gamma}^* \omega_A = \omega_A, \quad \hat{\gamma}^* \Omega_A = \delta_\gamma \Omega_A, \quad \delta_\gamma \in \{\pm 1\}, \quad (48)$$

where δ_γ is the product of the three complex signs in (47). On the flat end, the diagonal complex signs multiply the holomorphic volume form by δ_γ . Since both $\hat{\gamma}^*\Omega_A$ and $\delta_\gamma\Omega_A$ are parallel holomorphic volume forms with that same limit, they agree everywhere.

We denote the complex one-form on the tangent two-plane chosen above by $\nu = u^1 + iu^2$. With this choice, the product Cayley form is

$$\Phi_A = \frac{1}{2}\omega_A^2 + u^{12} \wedge \omega_A + \text{Re}(\nu \wedge \Omega_A). \quad (49)$$

For each normalizer element γ , its original affine derivative on $V_{A_r} \oplus W_{A_r}$ preserves Φ_0 . Together with $\gamma^*\Omega_{A_r} = \delta_\gamma\Omega_{A_r}$, the flat product identity forces its tangent action to satisfy

$$\gamma^*\nu = \delta_\gamma\nu. \quad (50)$$

We denote the resulting product lift on $V_{A_r} \times Y_A$ by

$$\tilde{\gamma} = (D\gamma|_{V_{A_r}}) \times \hat{\gamma}.$$

Substitution of (48) and (50) into (49) gives $\tilde{\gamma}^*\Phi_A = \Phi_A$ and $\tilde{\gamma}^*((u^1)^2 + (u^2)^2 + g_A) = (u^1)^2 + (u^2)^2 + g_A$ exactly. At a point, the two involutions with four-dimensional fixed space act as $(-I_U, I_{U'})$ and $(I_U, -I_{U'})$. We use the product $\text{EH}_\ell(U) \times \text{EH}_L(U')$ from (42). Because $B_p = 1$, condition (iii) imposes no nontrivial residual automorphism on the model attached to a fixed point component; it remains to define transports between distinct points in the same K -orbit. If $p' = \gamma p$, normality of G preserves the unique old subgroup $A_p \cap G$; hence $D\gamma$ carries the labelled character splitting $U_p \oplus U'_p$ to $U_{p'} \oplus U'_{p'}$ without interchanging the unequal-scale factors. We denote the radial lifts of the two linear quotient maps by

$$\hat{\gamma}_U : \text{EH}_\ell(U_p) \longrightarrow \text{EH}_\ell(U_{p'}), \quad \hat{\gamma}_{U'} : \text{EH}_L(U'_p) \longrightarrow \text{EH}_L(U'_{p'}).$$

We choose each lift so that its action on the Eguchi–Hanson hyperkähler triple is the orthogonal transformation induced by the corresponding restriction of $D\gamma$ on flat self-dual two-forms; this determines whether the lift is holomorphic or antiholomorphic. We define

$$\chi_{\gamma,p} = \hat{\gamma}_U \times \hat{\gamma}_{U'}. \quad (51)$$

Under these lifts, every term in the bilinear formula (43) transforms exactly as its flat-limit counterpart. Since $(D\gamma)^*\Phi_0 = \Phi_0$, it follows that $\chi_{\gamma,p}^*\Phi_{p'} = \Phi_p$, and the radial isometries also give $\chi_{\gamma,p}^*g_{p'} = g_p$.

For each two-stratum orbit, we define the data on a translate $C' = \gamma C_A$ by transporting the data on the chosen representative through γ . If two elements give the same translate, their quotient lies in $N_K(C_A)$, and the proved B_{C_A} -invariance shows that the transported datum is independent of the choice. We define the isolated-point data orbitwise by the same procedure.

Every transport map constructed above agrees, away from the exceptional locus, with the lift of the original affine action. Therefore, for every component C and every $\gamma, \delta \in K$, the two sides of

$$\chi_{\gamma\delta,C} = \chi_{\gamma,\delta C} \circ \chi_{\delta,C} \quad (52)$$

agree on the dense open set on which the blowdown is an isomorphism. Since both sides are continuous maps, equality on that dense set implies equality everywhere. Thus the transports satisfy Joyce's cocycle (52). Conditions (i)–(iii), and hence all the requirements of [10, Definition 13.2.2], are satisfied. $\square$

3.3 The global Joyce atlas and analytic gluing

We now specialize Joyce's Definition 13.2.5 to the full nested isotropy stratification of $\mathbb{T}^8/K$. We use $\mathcal{I}$ to index the distinct connected submanifolds that occur as components of $\text{Fix}(A)$ for subgroups $A \leq K$, together with the ambient component $F_0 = \mathbb{T}^8$. The bold index $\mathbf{0}$ is distinct from the two-stratum subgroup A_0 in (15). For each index, we define

$$A_i = C_K(F_i), \quad V_i = TF_i, \quad W_i = V_i^\perp, \quad B_i = N_K(F_i)/A_i.$$

The partial order $i \succeq j$ means $F_i \subset F_j$, so indices higher in this partial order correspond to deeper strata. Besides $i \succ \mathbf{0}$, the only strict chains consist of a two-stratum above one of its incident four-strata or an isolated point above one of its two incident four-strata, as displayed below:

$$\text{two-stratum} \succ \text{four-stratum} \succ \mathbf{0}, \quad \text{point} \succ \text{four-stratum} \succ \mathbf{0}.$$

We choose Joyce's dimension-indexed constants $\zeta_0, \zeta_1, \zeta_2, \zeta_4, \zeta_8 > 0$ and define $\xi_i = \zeta_{\dim V_i}$. Only dimensions 0, 2, 4, 8 occur. We set

$$O_i = \{x : d(x, F_i) \leq \xi_i\}, \quad P_i = \{x : d(x, F_i) < 2\xi_i\}, \quad Q_i = P_i \setminus \bigcup_{j \succ i} O_j. \quad (53)$$

We choose the constants in increasing order of $\dim V_i$, beginning with the zero-dimensional strata, and make each new constant sufficiently small that two incomparable neighborhoods P_i, P_j can meet only inside a neighborhood O_k of a component satisfying $k \succ i, j$. Such a choice exists because every intersection of two affine fixed flats is a finite union of components belonging to deeper fixed sets. Joyce's [10, Proposition 13.2.4] then proves

$$Q_i \cap Q_j \neq \emptyset \implies i \succeq j \text{ or } j \succeq i.$$

The same proposition supplies, for each i , a B_i -equivariant flat tubular identification of Q_i/A_i with an open subset of $F_i \times (W_i/A_i)$, and it proves that the B_i -action on that open set is free.

For each singular index i , we denote by $\pi_i : Y_i \rightarrow W_i/A_i$ the model appropriate to its stratum: an Eguchi–Hanson resolution in codimension four, Y_A in codimension six, and $\text{EH}_\ell \times \text{EH}_L$ at an isolated point. For $i = \mathbf{0}$, we take the identity model. We define

$$M_i = (\text{id}_{F_i} \times \pi_i)^{-1}(Q_i/A_i) \subset F_i \times Y_i. \quad (54)$$

For $i \succeq j$, we denote by $\psi'_{i,j}$ the local-product embedding constructed in the proof of Proposition 3.2, and we set $\psi'_{i,i} = \text{id}$. We now define the relation in (57) explicitly. If $x \in M_j$ and $y \in M_k$, declare $x \sim y$ precisely when one of the following two clauses holds:

1. There exists $\gamma \in K$ such that $i = \gamma \cdot k$ satisfies $i \succeq j$, such that $x \in M_j \cap (P_i \times Y_j)$, and

$$(\text{id} \times \psi'_{i,j})(x) = (\gamma \times \chi_{\gamma,k})(y); \quad (55)$$

2. There exists $\gamma \in K$ such that $i = \gamma \cdot j$ satisfies $i \succeq k$, such that $y \in M_k \cap (P_i \times Y_k)$, and

$$(\gamma \times \chi_{\gamma,j})(x) = (\text{id} \times \psi'_{i,k})(y). \quad (56)$$

In both clauses, $\gamma \times \chi_{\gamma,k} : F_k \times Y_k \rightarrow F_i \times Y_i$ acts by the affine map $\gamma : F_k \rightarrow F_i$ on the first factor and by the resolution transport on the second. These clauses reproduce [10, Definition 13.2.5]. The cocycle (52) and equality of the incidence transition maps on nested local-product charts imply

reflexivity, symmetry, and transitivity and identify the two composites on every triple overlap. Hence

$$\widehat{Z} = \left(\prod_{i \in \mathcal{I}} M_i \right) / \sim \quad (57)$$

is covered by the smooth patches M_i/B_i . The action of B_i on M_i is free because it covers the free action on Q_i/A_i . Thus (57) is a smooth eight-manifold. The maps $\text{id} \times \pi_i$ glue to a blowdown

$$\pi : \widehat{Z} \longrightarrow \mathbb{T}^8/K. \quad (58)$$

Properness can be checked locally on the target. We choose a finite refinement of the Joyce cover by relatively compact open sets U_α . For each $\overline{U}_\alpha$, its inverse image under π is a closed subset of the union of finitely many inverse images under the proper model maps $\text{id} \times \pi_i$, and is therefore compact. Thus π is proper. Since π is surjective and $\mathbb{T}^8/K$ is compact, $\widehat{Z}$ is compact. The atlas, equivalence relation, and blowdown just constructed are the objects of [10, Definition 13.2.5] in the present notation.

The scaled data entering the analytic theorem are as follows. For $t \in (0, 1]$, we retain the same Y_i and define

$$\pi_i^t = t\pi_i, \quad \Phi_i^t = t^4 D_{t;i}^* \Phi_i, \quad g_i^t = t^2 D_{t;i}^* g_i,$$

where

$$D_{t;i} : V_i \times Y_i \longrightarrow V_i \times Y_i, \quad D_{t;i}(v, y) = (t^{-1}v, y).$$

Here $t\pi_i$ means postcomposition of π_i with radial dilation by t on the cone W_i/A_i . Thus the flat metric along V_i is unchanged and the resolving normal metric is scaled by t^2 . We define

$$M_i^t = (\text{id}_{F_i} \times \pi_i^t)^{-1}(Q_i/A_i).$$

For every strict incidence $i \succ j$, we define

$$V'_{ij} = V_j \cap W_i, \quad V_j = V_i \oplus V'_{ij}.$$

Condition (ii) gives an equivariant QALE identification $\psi'_{i,j} : \mathcal{U}'_{ij} \rightarrow Y_i$ on an open overlap $\mathcal{U}'_{ij} \subset V'_{ij} \times Y_j$. We define

$$\mathcal{U}_{ij} = V_i \times \mathcal{U}'_{ij} \subset V_i \times V'_{ij} \times Y_j = V_j \times Y_j, \quad \Psi_{ij} = \text{id}_{V_i} \times \psi'_{i,j}.$$

The displayed product decomposition identifies $\mathcal{U}_{ij}$ with an eight-dimensional open subset of $V_j \times Y_j$. Proposition 13.1.9 of [10] supplies an equivariant $\sigma_{ij} \in \Omega^3(\mathcal{U}_{ij})$ satisfying

$$d\sigma_{ij} = \Psi_{ij}^* \Phi_i - \Phi_j. \quad (59)$$

We define

$$D'_{t;ij} : V'_{ij} \times Y_j \longrightarrow V'_{ij} \times Y_j, \quad D'_{t;ij}(v', y) = (t^{-1}v', y), \\ \mathcal{U}_{ij}^{t,t} = (D'_{t;ij})^{-1}(\mathcal{U}'_{ij}), \quad \mathcal{U}_{ij}^t = D_{t;j}^{-1}(\mathcal{U}_{ij}) = V_i \times \mathcal{U}_{ij}^{t,t},$$

and we define

$$(\psi'_{i,j})^t = \psi'_{i,j} \circ D'_{t;ij} : \mathcal{U}_{ij}^{t,t} \longrightarrow Y_i, \quad \Psi_{ij}^t = \text{id}_{V_i} \times (\psi'_{i,j})^t = D_{t;i}^{-1} \circ \Psi_{ij} \circ D_{t;j}, \quad \sigma_{ij}^t = t^4 D_{t;j}^* \sigma_{ij}.$$

We use these scaled local-product maps and the unchanged equivariant transports $\chi_{\gamma,i}^t = \chi_{\gamma,i}$ in the two clauses (55)–(56), with M_i replaced by M_i^t . The quotient defined by those clauses is denoted by $\widehat{Z}^t$. The local maps $\text{id} \times \pi_i^t$ agree under the equivalence relation and therefore define

$$\pi^t : \widehat{Z}^t \longrightarrow \mathbb{T}^8/K. \quad (60)$$

The local models and all transition maps depend smoothly on t . The parameter-dependent patch charts and the graphs of their transition maps therefore give the total quotient over $(0, 1]$ the structure of a smooth manifold, and projection to t is a submersion with fibre $\widehat{Z}^t$. Over a compact subinterval $J \subset (0, 1]$, the properness argument used for (58), applied to the finite parameterized patch cover, shows that the inverse image of J is compact. Ehresmann's theorem then makes the restriction over J a smooth fibre bundle. These restrictions are compatible as J varies, so the total quotient is a smooth fibre bundle over $(0, 1]$. Since this interval is contractible, the bundle admits a smooth trivialization. We fix one such trivialization $\vartheta_t : \widehat{Z} \rightarrow \widehat{Z}^t$ with $\vartheta_1 = \text{id}$ and pull all subsequent forms and maps back by ϑ_t . We retain the same symbols for the transported objects and also write M_i^t for the inverse-image patches in the fixed manifold $\widehat{Z}$. This gives the smooth connected family of scaled resolutions in [10, Definition 13.4.1], written here with an explicit fixed fibre.

The conjugation formula for Ψ_{ij}^t is forced by the scaled blowdowns. Under the fixed identification $W_i/A_i \cong V'_{ij} \times (W_j/A_j)$, it gives

$$\pi_i^t((\psi'_{i,j})^t(v', y)) = (v', \pi_j^t(y)).$$

Applying $t^4 D_{t,j}^*$ to (59), and using the conjugation formula, gives

$$d\sigma_{ij}^t = (\Psi_{ij}^t)^* \Phi_i^t - \Phi_j^t, \quad (61)$$

and the scaled forms retain the full derivative estimates in Proposition 13.1.9 of [10]. We choose the unscaled primitives simultaneously, as in [10, Section 13.3.1], so that

$$(\Psi_{jk})^* \sigma_{ij} + \sigma_{jk} - \sigma_{ik} = 0 \quad (i \succ j \succ k),$$

on every triple overlap. Scaling this identity gives

$$(\Psi_{jk}^t)^* \sigma_{ij}^t + \sigma_{jk}^t - \sigma_{ik}^t = 0 \quad (i \succ j \succ k),$$

which is the additive cocycle needed on the scaled triple overlaps.

We now specify the correction terms. For every four-dimensional component F_j , Section 13.3.2 of [10] defines a homogeneous three-form τ_j on $\mathbb{R}^8 \setminus V_j$ as the leading homogeneous term of σ_{j0} . If $i \succeq j$, the R -data overlap map and the flat tubular quotient push τ_j down to an invariant three-form on $Q_i \setminus F_j$. The pushed-down forms obtained from two comparable indices coincide on their common domain. We choose the special partition of unity in [10, Definition 13.2.6] and denote its functions by η'_i . These functions satisfy, simultaneously,

$$\text{supp } \eta'_i \subset Q_i, \quad \eta'_{\gamma i} \circ \gamma = \eta'_i, \quad \sum_{i \in \mathcal{I}} \eta'_i = 1,$$

and the following normal-direction condition. For every $i \neq \mathbf{0}$ and every j , there is a smooth function $\bar{\eta}_{ij}$ on F_i such that, on a neighborhood of F_i ,

$$\eta'_j = \text{pr}_{F_i}^* \bar{\eta}_{ij}.$$

Thus every partition function, not only η'_i , is constant in the W_i -directions there. For a fixed index j of dimension four, we define, initially on the complement of F_j ,

$$\Xi_j = \sum_{i \succeq j} d\eta'_i \wedge \tau_j = d \left(\sum_{i \succeq j} \eta'_i \right) \wedge \tau_j. \quad (62)$$

Near F_j , every partition function that can be nonzero has an index $i \succeq j$. Consequently the parenthesized sum in (62) is identically one on a neighborhood of F_j , and its differential is identically zero there. Therefore Ξ_j is zero on a punctured neighborhood of F_j and has a unique smooth extension across F_j obtained by setting it equal to zero. The pushed-down forms agree on overlaps, so

$$\Xi = \sum_{\dim V_j=4} \Xi_j$$

is a globally defined smooth K -invariant four-form on $\mathbb{T}^8$. We use the same symbol for the descended orbifold four-form on $\mathbb{T}^8/K$. The flat metric and its Hodge star are K -invariant, so the Hodge decomposition restricts to the K -invariant de Rham complex of $\mathbb{T}^8$, equivalently to the orbifold de Rham complex of $\mathbb{T}^8/K$. Applied to $d\Xi$, this decomposition first gives an invariant anti-self-dual four-form Φ_0^- satisfying $d\Phi_0^- = d\Xi$. Any two such anti-self-dual solutions differ by a harmonic anti-self-dual four-form. The difference $\Phi_0^- - \Xi$ is closed. There is therefore a unique harmonic anti-self-dual four-form h^- for which

$$[\Phi_0^- + h^- - \Xi] \in H_+^4(\mathbb{T}^8/K; \mathbb{R}).$$

Indeed, h^- is the negative of the anti-self-dual harmonic component of $[\Phi_0^- - \Xi]$. We set $\Phi^- = \Phi_0^- + h^-$. This proves both existence and uniqueness of the anti-self-dual solution for which $[\Phi^- - \Xi]$ is self-dual. We denote the harmonic self-dual representative of this class by Ψ . The closed form $\Phi^- - \Xi - \Psi$ represents zero in cohomology and is therefore exact. We choose an invariant three-form Υ satisfying

$$d\Upsilon = \Phi^- - \Xi - \Psi. \quad (63)$$

These are exactly the forms constructed in [10, Section 13.3.3].

For an orbit $[i] \in \mathcal{I}/K$, we define $\eta'_{[i]} = \sum_{j \in [i]} \eta'_j$. This function is K -invariant and therefore descends to $\mathbb{T}^8/K$. We then define

$$\eta_{[i]}^t = (\pi^t)^* \eta'_{[i]}.$$

The normal-direction condition implies that this pullback is smooth across every exceptional set. The functions $\eta_{[i]}^t$ form a partition of unity on the fixed manifold $\hat{Z}$. We fix an index i . In a summand indexed by an orbit $[j]$, the relation $j \succ i$ (respectively, $i \succ j$) means that one has chosen a representative $j \in [j]$ comparable with i . The K -action and the cocycle (52) identify the local forms obtained from any two such representatives. Consequently each orbit sum below is independent of this choice and descends to M_i^t/B_i . On that quotient patch, we define the pregluing form by

$$\xi^t = \Phi_i^t + \sum_{\substack{[j] \in \mathcal{I}/K \\ j \succ i}} d(\eta_{[j]}^t \sigma_{ji}^t) - \sum_{\substack{[j] \in \mathcal{I}/K \\ i \succ j}} d(\eta_{[j]}^t \sigma_{ij}^t) + t^4 d(f^t(\pi^t)^* \Upsilon) + t^4 \Psi^t. \quad (64)$$

The first two sums correct the exact differences (61). In the first sum every product $\eta_{[j]}^t \sigma_{ji}^t$, and in the second every product $\eta_{[j]}^t \sigma_{ij}^t$, is extended by zero outside the corresponding orbit patch. The

support and normal-constancy conditions make both kinds of extension smooth; equivariance gives the representative-independence asserted above.

It remains to define f^t and Ψ^t . We denote the singular set by $S \subset \mathbb{T}^8/K$ and the image of F_i by $\overline{F}_i$. We also define

$$\rho_t(x) = [t + d(\pi^t(x), S)]^{-1}.$$

This is the weight function of [10, Definition 13.4.2]; the weighted estimates in [10, Propositions 13.4.3–13.5.7] are all taken with respect to ρ_t . Using the radii in (53), Definition 13.4.2 gives the open subsets

$$\begin{aligned} D^t &= \{x : d(\pi^t(x), \overline{F}_i) > t^{1/3}\xi_i \text{ for every } i \neq \mathbf{0}\}, \\ E^t &= \{x : d(\pi^t(x), \overline{F}_i) > 2t^{1/3}\xi_i \text{ for every } i \neq \mathbf{0}\}. \end{aligned}$$

The inclusion $E^t \subset D^t$ holds, and both sets lie over the regular locus. Definition 13.5.1 supplies a smooth function f^t that is identically zero on a neighborhood of $\widehat{Z} \setminus D^t$, identically one on E^t , and satisfies

$$|\nabla^q f^t|_{g_i^t} = O(t^{-q/3}) \quad \text{on } (M_i^t/B_i) \cap (D^t \setminus E^t), \quad q = 0, 1, 2, \quad i \in \mathcal{I}.$$

The form $f^t(\pi^t)^*\Upsilon$ in (64) is evaluated on the regular support of f^t and extended by zero; it is smooth because the cutoff vanishes on a neighborhood of the complement of D^t . The form Ψ^t is the closed extension furnished by [10, Proposition 13.4.4]; it equals $(\pi^t)^*\Psi$ on D^t and $[\Psi^t] = (\pi^t)^*[\Psi]$. Equation (63) cancels the leading homogeneous t^4 contribution from every four-dimensional stratum. On an overlap, equation (61) and the additive cocycle for the σ_{ij}^t show that the two local expressions in (64) agree. They therefore define a global four-form ξ^t . Each local expression is closed because $d\Phi_i^t = 0$, $d^2 = 0$, and $d\Psi^t = 0$. Hence $d\xi^t = 0$. This is [10, Definition 13.5.1 and Proposition 13.5.2].

Let $\mathcal{A}\widehat{Z} \subset \Lambda^4 T^* \widehat{Z}$ be the bundle of admissible four-forms. Definition 10.5.8 of [10] constructs an open tubular neighborhood $\mathcal{T}\widehat{Z} \subset \Lambda^4 T^* \widehat{Z}$ and a specific smooth fibre map $\Theta : \mathcal{T}\widehat{Z} \rightarrow \mathcal{A}\widehat{Z}$. Pointwise, each element of $\mathcal{T}\widehat{Z}$ has a unique expression $\Omega + \chi$, where Ω is admissible, $\chi \in \Lambda_{27}^4$ for the metric induced by Ω , and $|\chi|$ is smaller than the fixed tubular radius; Joyce defines $\Theta(\Omega + \chi) = \Omega$. To avoid confusion with the holomorphic volume form used earlier, we denote this particular map by Θ_{adm} below. Proposition 10.5.9 gives its nonlinear expansion and estimates. For all sufficiently small t , the form ξ^t takes values in the domain of Θ_{adm} . We define

$$\Phi_{\text{pre}}^t = \Theta_{\text{adm}}(\xi^t), \quad \phi^t = \xi^t - \Phi_{\text{pre}}^t, \quad g^t = g_{\Phi_{\text{pre}}^t}. \quad (65)$$

Then

$$d\Phi_{\text{pre}}^t + d\phi^t = 0. \quad (66)$$

The cancellation in (64) removes the leading codimension-four torsion term. The resulting estimates in [10, Theorem 13.5.8] give constants independent of t such that

$$\|\phi^t\|_{L^2} \leq Ct^{13/3}, \quad \|d\phi^t\|_{L^{10}} \leq Ct^{7/5}, \quad \delta(g^t) \geq ct, \quad \|R(g^t)\|_{C^0} \leq Ct^{-2}. \quad (67)$$

Theorem 3.3. *There exists $\varepsilon > 0$ such that the compact smooth eight-manifold $\widehat{Z}$ carries a torsion-free $\text{Spin}(7)$ structure $\widehat{\Phi}_t$ for every $t \in (0, \varepsilon)$.*

Proof. Equations (66)–(67) are exactly the hypotheses of [10, Theorem 13.6.1]. In the notation of its proof, there is a small anti-self-dual four-form η^t satisfying

$$d\eta^t = d\phi^t + dF_{\Phi_{\text{pre}}^t}(\eta^t), \quad \|\eta^t\|_{C^0} = O(t^{1/3}),$$

where the quadratic normal correction is characterized by

$$\Theta_{\text{adm}}(\Phi_{\text{pre}}^t + \eta) = \Phi_{\text{pre}}^t + \eta - F_{\Phi_{\text{pre}}^t}(\eta).$$

Consequently,

$$\widehat{\Phi}_t = \Theta_{\text{adm}}(\Phi_{\text{pre}}^t + \eta^t) = \Phi_{\text{pre}}^t + \eta^t - F_{\Phi_{\text{pre}}^t}(\eta^t)$$

is an admissible four-form. Equations (66) and the nonlinear equation give the explicit calculation

$$\begin{aligned} d\widehat{\Phi}_t &= d\Phi_{\text{pre}}^t + d\eta^t - dF_{\Phi_{\text{pre}}^t}(\eta^t) \\ &= -d\phi^t + (d\phi^t + dF_{\Phi_{\text{pre}}^t}(\eta^t)) - dF_{\Phi_{\text{pre}}^t}(\eta^t) = 0. \end{aligned}$$

An admissible four-form defines a $\text{Spin}(7)$ structure, and such a structure is torsion-free if and only if its defining four-form is closed. Thus $\widehat{\Phi}_t$ is torsion-free. This is the existence conclusion of [10, Theorem 13.6.2]. $\square$

4 The equivariant first resolution and the global blowdown

The Joyce resolution admits a topological description by two successive fibrewise resolutions. This description identifies its cohomology and fundamental group in Sections 5 and 6.

4.1 The Martín–Merchán resolution

The singular locus of X is the disjoint union of ten associative three-manifolds $N_1, \dots, N_{10}$ of exact normal isotropy $\mathbb{Z}_2$ [11, Section 2]. Using the component frames, radii, and lifted transition maps already fixed for the old G -strata in the Joyce atlas, we denote the corresponding first-stage resolution by

$$r : M \longrightarrow X. \tag{68}$$

The relative equivariant tubular-neighborhood argument at the end of Lemma 4.1 identifies it, over X and relative to the exterior, with the resolution used in [11]. It replaces each normal $\mathbb{C}^2/\{\pm 1\}$ by

$$Y = \{(w, \ell) \in \mathbb{C}^2 \times \mathbb{CP}^1 : w \in \ell\} / ((w, \ell) \sim (-w, \ell)). \tag{69}$$

The fibrewise squaring map identifies this smooth complex surface with $\text{Tot } \mathcal{O}_{\mathbb{CP}^1}(-2)$, the minimal resolution of $\mathbb{C}^2/\{\pm 1\}$. The manifold M is simply connected [11, Section 2.2]. We write $E_j = r^{-1}(N_j)$ and denote its exceptional Thom class by $x_j \in H^2(M)$. The normal complex structure is defined by

$$g(I_j v, w) = \varphi(\chi_j, v, w), \quad \chi_j = \begin{cases} \partial_{y_3}, & j = 1, 2, \\ \partial_{x_3}, & j \geq 3. \end{cases} \tag{70}$$

Lemma 4.1. *The involution τ lifts to an involution $\tilde{\tau} : M \rightarrow M$ with $r \circ \tilde{\tau} = \tau \circ r$, and*

$$\tilde{\tau}^* x_1 = -x_2, \quad \tilde{\tau}^* x_2 = -x_1, \quad \tilde{\tau}^* x_j = x_j \quad (3 \leq j \leq 10). \tag{71}$$

Proof. At $t = 0$, we express the action in the same mapping-torus chart by defining

$$\tau_0 = F^{-1}\tau, \quad \tau_0(z_1, z_2, z_3) = (-iz_1, -iz_2, z_3 - \frac{1}{4}). \tag{72}$$

At $t = 0$ the two relevant chosen covers are

$$\begin{aligned} N_1^1 &= \{(x_1 - ix_1, x_2 - ix_2, iy_3) : x_1, x_2, y_3 \in \mathbb{R}/\mathbb{Z}\}, \\ N_2^1 &= \{(x_1 + ix_1, x_2 + ix_2, -\frac{1}{4} + iy_3) : x_1, x_2, y_3 \in \mathbb{R}/\mathbb{Z}\}. \end{aligned}$$

Consequently τ_0 sends the displayed point of N_1^1 to

$$(-x_1 - ix_1, -x_2 - ix_2, -\frac{1}{4} + iy_3) \in N_2^1.$$

The same substitution on N_2^1 gives

$$\tau_0(N_1^1) = N_2^1, \quad \tau_0(N_2^1) = N_1^2 = \iota_1 N_1^1.$$

At $t = \frac{1}{2}$, the covers of $N_3, \dots, N_6$ have fixed imaginary half-periods in the three elliptic coordinates, and τ preserves each chosen component. The covers of $N_7, \dots, N_{10}$ have fixed real half-periods in z_1, z_2 and a fixed imaginary quarter-period in z_3 ; $\iota_2 \tau$ preserves each chosen component. Thus the maps on the component covers are

components	map on the component cover	deck involution
$N_1^1 \longrightarrow N_2^1$	τ_0	ι_2
$N_2^1 \longrightarrow N_1^1$	$\iota_1 \tau_0$	ι_2
$N_j^1 \longrightarrow N_j^1, \ 3 \leq j \leq 6$	τ	ι_1
$N_j^1 \longrightarrow N_j^1, \ 7 \leq j \leq 10$	$\iota_2 \tau$	$\iota_1.$

(73)

Here τ_0 commutes with ι_1, ι_2 and $\tau_0^2 = \iota_1$. Thus the first two displayed maps are inverse to one another and are ι_2 -equivariant. The maps in the last two rows square to the identity and commute with ι_1 .

Every representative L in (73) satisfies $L^* \varphi = -\varphi$. If $L_* \chi_j = \chi_k$, then for normal vectors v, w ,

$$g(I_k L v, L w) = \varphi(\chi_k, L v, L w) = -\varphi(\chi_j, v, w) = g(-L I_j v, L w).$$

Thus $I_k L = -L I_j$. In particular, $D\tau_0 \chi_1 = \chi_2$ gives

$$D\tau_0 I_1 = -I_2 D\tau_0. \tag{74}$$

The same identity holds for $\iota_1 \tau_0$. In the last two rows of (73), the derivative sends χ_j to $-\chi_j$; the same displayed calculation now has two minus signs, which cancel, and

$$D\tau I_j = I_j D\tau \quad (j \geq 3). \tag{75}$$

Here $D\tau$ denotes the derivative of the representative in the relevant row; the translations ι_2 do not alter the calculation.

We now construct the global lift, including the normal orbifold quotients. In the notation of Martín-Merchán, the resolved local piece over N_j is

$$\mathcal{P}_j = (N_j^1 \times Y) / \langle d_j \rangle, \quad d_j = \begin{cases} \iota_2, & j = 1, 2, \\ \iota_1, & 3 \leq j \leq 10. \end{cases} \tag{76}$$

In the unitary normal frame used in (70), the deck map acts on the two complex normal coordinates by $\text{diag}(1, -1)$; the central $-I$ has already been divided out in the definition of Y . Thus

$$d_j(x, [(w, \lambda)]) = (d_j x, [(Dd_j w, Dd_j \lambda)]).$$

A linear or antilinear isometry L lifts canonically to Y by

$$[(w, \lambda)] \longmapsto [(Lw, L\lambda)]. \tag{77}$$

We define $L_1 = \tau_0 : N_1^1 \rightarrow N_2^1$ and $L_2 = \iota_1 \tau_0 : N_2^1 \rightarrow N_1^1$. We also define $L_j = \tau$ for $3 \leq j \leq 6$ and $L_j = \iota_2 \tau$ for $7 \leq j \leq 10$. The commutation relations in (73) hold on both the base and the normal frame, so

$$L_j d_j = d_k L_j \quad (78)$$

when L_j maps the j th component to the k th. Hence

$$\widehat{L}_j[x, y] = [L_j x, \widehat{D} L_j y]$$

is a well-defined map between the quotients (76). The compositions are exact:

$$(\iota_1 \tau_0) \tau_0 = \iota_1 \tau_0^2 = 1, \quad \tau_0(\iota_1 \tau_0) = 1, \quad \tau^2 = (\iota_2 \tau)^2 = 1.$$

Thus the lifted local maps satisfy the involution cocycle before gluing.

For the tubular gluing, we choose an ambient metric on a neighborhood of the disjoint union of the lifted strata in the smooth mapping-torus cover and average it over the finite group generated by the old deck maps and the L_j . The resulting orthogonal normal bundles and ambient exponential maps are equivariant. We choose the radii and radial cutoff functions to be constant on each orbit of components. On the common annulus, (77) agrees with the original linear action through the blowdown, so it glues to that action on the complement. The maps $\widehat{L}_j$ therefore define a global involution $\tilde{\tau}$ with $r \circ \tilde{\tau} = \tau \circ r$.

Finally, this equivariant choice has the same underlying resolution as the one used in [11]. Indeed, the equivariant tubular-neighborhood uniqueness theorem [4, Chapter VI] gives an equivariant isotopy between any two such tubular embeddings after shrinking the radii; isotopy extension [9, Chapter 8] gives a diffeomorphism relative to the complement of the resolution neighborhoods. Hence the exceptional labels and the cohomology-ring calculation from [11] apply to the present r . For $j = 1, 2$ the lift is antiunitary and conjugates the $\mathcal{O}(-2)$ fibre coordinate, so it reverses the Thom orientation. For $j \geq 3$ it is unitary and preserves that orientation. This proves (71). $\square$

The sign in (27) is equivalently determined by $D\gamma I_C = \epsilon_{\gamma, C} I_{\gamma C} D\gamma$. Thus its cocycle law is automatic. In particular $\epsilon_{\tau, N_1} = -1$, while it is $+1$ on N_j , $j \geq 3$; the R -data uses the same exceptional orientations.

4.2 The second stage

We define

$$W = S^1 \times M, \quad \sigma(s, p) = (-s, \tilde{\tau} p), \quad Z_0 = W / \langle \sigma \rangle. \quad (79)$$

Because τ normalizes G , it induces $\bar{\tau} : X \rightarrow X$, and there is a canonical identification

$$\mathcal{O} = (S^1 \times \mathbb{T}^7) / (G \rtimes \langle T \rangle) \cong (S^1 \times X) / \langle \bar{T} \rangle, \quad \bar{T}(s, x) = (-s, \bar{\tau} x).$$

The identity $(\text{id}_{S^1} \times r) \circ \sigma = \bar{T} \circ (\text{id}_{S^1} \times r)$ makes $\text{id}_{S^1} \times r$ descend to a blowdown

$$q_0 : Z_0 \longrightarrow \mathcal{O}, \quad q_0([s, p]) = [s, r(p)]. \quad (80)$$

It is proper because its domain is compact.

Lemma 4.2. *The fixed locus of σ is a disjoint union of smooth four-manifolds. Resolving the resulting A_1 singularities gives a smooth manifold $\widehat{Z}_{\text{st}}$ and a proper degree-one map*

$$\rho : \widehat{Z}_{\text{st}} \longrightarrow Z_0. \quad (81)$$

This staged resolution is diffeomorphic, compatibly over the regular locus, to the Joyce resolution $\widehat{Z}$.

Proof. Since σ is a smooth involution, its fixed locus is a smooth embedded submanifold. Away from the old exceptional set, the exact-isotropy part of Table 1 says that all its branches have dimension four. Every zero-dimensional branch lies on an old stratum by (14). There are two local intersection models.

At a codimension-six stratum, we choose coordinates in which

$$\begin{aligned} a(u, z_1, z_2, z_3) &= (u, z_1, -z_2, -z_3), \\ h(u, z_1, z_2, z_3) &= (u, -z_1, z_2, -z_3). \end{aligned} \tag{82}$$

We resolve the old a -quotient in (z_2, z_3) by the scale- L Eguchi–Hanson model Y . The residual map is induced by

$$(z_1, z_2, z_3) \mapsto (-z_1, z_2, -z_3).$$

On Y , the lift of $(z_2, z_3) \mapsto (z_2, -z_3)$ has two disjoint fixed complex lines, over $[1 : 0]$ and $[0 : 1]$ in the exceptional $\mathbb{CP}^1$. More explicitly, if

$$L_2 = \{[(w_2, 0), [1 : 0]] : w_2 \in \mathbb{C}\}, \quad L_3 = \{[(0, w_3), [0 : 1]] : w_3 \in \mathbb{C}\},$$

then the residual fixed set is

$$\mathbb{R}_u^2 \times \{0\}_{z_1} \times (L_2 \sqcup L_3).$$

It consists of two disjoint real four-dimensional branches. We resolve both branches. This operation inserts the remaining two midpoint rays with scale ℓ and gives the staged fan (38) and the support data (40).

At an isolated product point, we choose the representation $a = (I_U, -I_{U'})$ and $h = (-I_U, -I_{U'})$. After resolving the a -quotient, the model is

$$U \times \text{EH}_L(U'),$$

and the residual involution is $(-I_U, \text{id}_{\text{EH}_L(U')})$. Its fixed set is $\{0\} \times \text{EH}_L(U')$, again of real dimension four. The second resolution is $\text{EH}_\ell(U) \times \text{EH}_L(U')$ as in (42).

These models are exhaustive for the following group-theoretic reason. If $p \in \text{Fix}(\sigma)$ lies over the old exceptional set, then $q_0([p]) \in \mathcal{O}$ has stabilizer containing a nonidentity old element $a \in G$ and a new element $h \in GT$. Proposition 2.4 says that $\langle a, h \rangle$ is either a codimension-six $\mathbb{Z}_2^2$ with fixed-dimension pattern $(4, 4, 4)$, or an isolated $\mathbb{Z}_2^2$ with pattern $(4, 4, 0)$; no larger stabilizer occurs. The first case is exactly the $L_2 \sqcup L_3$ calculation and the second is exactly $\{0\} \times \text{EH}_L(U')$. Points away from the old exceptional set have exact new $\mathbb{Z}_2$ isotropy and give the regular four-dimensional branches already identified from Table 1. Thus every component B of $\text{Fix}(\sigma)$ has codimension four.

We now define the fibrewise resolution globally. For each component B of $\text{Fix}(\sigma)$, we identify B with its image in Z_0 and write $\nu B = TW|_B/TB$ for its normal bundle. The derivative of σ is the identity on TB and minus the identity on νB .

On the dense regular locus, $\tilde{\tau}$ agrees with τ , whose linear part has determinant -1 ; hence $\tilde{\tau}$ reverses the orientation of M . Reflection reverses the orientation of S^1 , so σ preserves the product orientation of W . Each tangent four-plane of B has the Cayley orientation in its local product model, and the transition maps preserve the corresponding Cayley form. These local orientations give a global orientation of TB . We orient νB so that the direct-sum orientation of $TB \oplus \nu B$ is the ambient orientation. In each R -data chart, the chosen normal frame identifies the normal fibre with the standard oriented $\mathbb{R}^4 = \mathbb{C}^2$. Let

$$H_B = U(2) \rtimes \langle \text{complex conjugation} \rangle \subset SO(4).$$

The normalizer matrices (22) and (47), together with the cocycle (52), show that every change between these normal frames takes values in H_B and that the changes satisfy the cocycle identity. In a codimension-six chart, the two residual branches are L_2, L_3 , and (36) identifies the normal-frame reduction obtained from the regular chart with the one obtained from the $\mathbb{C} \times \text{EH}_L$ chart. At an isolated chart, the two factors in (42) give the same identification. Consequently the normal frame cocycle defines a principal H_B -reduction P_B of the oriented linear frame bundle of νB .

The standard inner product on $\mathbb{R}^4$ is H_B -invariant, so P_B defines a metric on νB . The standard Kähler form is self-dual for this metric; $U(2)$ fixes it and complex conjugation negates it. Hence its unoriented span defines a smooth line bundle

$$\ell_B = P_B \times_{H_B} \mathbb{R}\omega_{\text{std}} \subset \Lambda^+ \nu^* B.$$

This order of construction is essential: the H_B -reduction first defines the compatible normal metric, and self-duality is asserted only for that metric. We extend the metric from $TB \oplus \nu B$ to a σ -invariant metric near B and then to W ; averaging does not alter its prescribed value along B because $D\sigma$ acts as $I \oplus (-I)$. The equivariant exponential map now identifies a neighborhood of B in Z_0 with the orbibundle

$$\nu B / \{\pm 1\} \longrightarrow B, \quad (83)$$

whose fibre is $\mathbb{R}^4 / \{\pm 1\}$.

Every transition in H_B has the canonical radial holomorphic or antiholomorphic lift to EH_ℓ constructed in (27); equation (52) says that these lifted transition functions satisfy the cocycle. Using the principal H_B -reduction P_B , we define

$$\mathcal{E}_B = P_B \times_{H_B} \text{EH}_\ell, \quad \rho_B([f, y]) = [f, \pi_{\text{EH}}(y)] \in P_B \times_{H_B} (\mathbb{R}^4 / \{\pm 1\}). \quad (84)$$

Here the notation means that H_B acts through its canonical lifted action on EH_ℓ ; equivalently, (84) is defined directly by the lifted transition cocycle. We truncate both bundles at the common chosen radius. On the boundary annulus the Eguchi–Hanson blowdown is the original quotient identification, so $\mathcal{E}_B$ glues to the unchanged exterior by the same fixed boundary identification. Performing this construction on the disjoint components defines the global map ρ in (81). It is an orientation-preserving diffeomorphism over the regular set; a regular value has one positive preimage, so $\deg \rho = 1$.

It remains to compare the gluing with the R -data resolution. The map r was defined using the restrictions of the Joyce component frames, tubular maps, radii, and lifted cocycle to the old subgroup G . At a two-stratum, the first blowup is the toric star subdivision by m_{23} in N_a . Passing to the residual quotient enlarges N_a to N ; the two nonsaturated boundary faces are $\langle e_1, e_2 \rangle$ and $\langle e_1, e_3 \rangle$, and their star subdivisions insert m_{12} and m_{13} . Hence the resulting fan is exactly (38), including its diagonal $\langle e_1, m_{23} \rangle$ and the support values corresponding to periods L, ℓ, ℓ . Every transition derivative lies in the relevant normalizer after the labels are transported from A_0 or A_1 . The exact calculation (46)–(47) shows that it fixes every ray label; in particular it preserves the fan and support function. Its toric lift is exactly the lift used in the R -data atlas. At a four-stratum the analogous statement is uniqueness of the lift to the minimal A_1 resolution, and at a zero-stratum it is the product lift to $\text{EH}_\ell(U) \times \text{EH}_L(U')$. Thus, for this chosen representative of the first-stage resolution, the staged atlas and the R -data atlas have the same local models, tubular maps, and lifted transition cocycle. In a codimension-six chart the composite “ m_{23} subdivision, residual lattice quotient, m_{12}, m_{13} subdivisions” is the toric transition associated with the fan (38); at a point it is the product blowdown (42). On every boundary annulus both use the original affine quotient map. Hence the transition maps coincide and glue to a global diffeomorphism over $\mathcal{O}$ whose restriction to the regular locus is the identity.

This identification does not depend on the chosen presentation of the first resolution. After the tubular radii are decreased if necessary, any two equivariant tubular embeddings of the same normal orbibundles are equivariantly isotopic. The isotopy lifts through the fixed local resolution model and extends across the common boundary annulus. If the strata are treated in decreasing order of depth, the isotopy extension theorem produces at each stage a diffeomorphism relative to the pieces already treated and relative to the exterior. This relative uniqueness identifies the chosen map r with the resolution map of any Martín–Merchán presentation while preserving the exceptional labels and the cohomology classes used below. $\square$

We henceforth identify $\widehat{Z}_{\text{st}}$ with $\widehat{Z}$ by the diffeomorphism constructed in Lemma 4.2.

5 A Massey product outside its full indeterminacy

We define

$$a = x_1 + x_2, \quad b = x_7 + x_3, \quad c = x_7 - x_3 \in H^2(M). \quad (85)$$

We use the following set-valued convention. If closed forms $\alpha_1, \alpha_2, \alpha_3$ of degrees p, q, r represent a_1, a_2, a_3 , and

$$d\xi_{12} = \alpha_1\alpha_2, \quad d\xi_{23} = \alpha_2\alpha_3,$$

then

$$\langle a_1, a_2, a_3 \rangle = [\xi_{12}\alpha_3 - (-1)^p\alpha_1\xi_{23}] + I(a_1, a_3), \quad (86)$$

where the full indeterminacy is

$$I(a_1, a_3) = a_1 H^{q+r-1} + H^{p+q-1} a_3 \quad (87)$$

in the cohomology of the ambient CDGA. As the representatives and primitives vary, the resulting cohomology classes form precisely the affine coset in (86). Consequently zero belongs to the Massey product if and only if one, and hence every, defining-system representative belongs to the full indeterminacy subspace (87).

Proposition 5.1 (Martín–Merchán). *The product $\langle a, b, c \rangle$ is defined and*

$$0 \notin \langle a, b, c \rangle. \quad (88)$$

Equivalently, every defining-system representative $Y \in H^5(M)$ satisfies

$$Y \notin I_M, \quad I_M = aH^3(M) + H^3(M)c. \quad (89)$$

Full-indeterminacy calculation. The additive splitting and products from [11, Section 3] are

$$H^k(M) = H^k(X) \oplus \bigoplus_{j=1}^{10} H^{k-2}(N_j)x_j, \quad (90)$$

$$x_j x_k = 0 \ (j \neq k), \quad x_j^2 = -2\text{PD}[N_j]. \quad (91)$$

For completeness, these formulas follow directly from the local replacement. We denote by U_0 the complement of smaller disjoint tubular neighborhoods of the N_j . Both X and M are obtained from U_0 by attaching bundles with the same boundary $N_j \widetilde{\times} \mathbb{RP}^3$; the singular fibre is $C(\mathbb{RP}^3)$ and the

resolved fibre is the truncated disc bundle of $\mathcal{O}_{\mathbb{CP}^1}(-2)$. Over $\mathbb{R}$, their fibre-pair cohomologies are, explicitly,

$$H^q(C(\mathbb{RP}^3), \mathbb{RP}^3; \mathbb{R}) = \begin{cases} \mathbb{R}, & q = 4, \\ 0, & q \neq 4, \end{cases} \quad H^q(D(\mathcal{O}(-2)), S(\mathcal{O}(-2)); \mathbb{R}) = \begin{cases} \mathbb{R}, & q = 2, 4, \\ 0, & q \neq 2, 4. \end{cases}$$

We spell out the bundle argument that turns this fibre calculation into a calculation of the relative groups. For each j , let T_j be the relative orientation class of the singular cone bundle, normalized to have fibre integral one, and choose a closed, compactly supported Thom form θ_j of the exceptional divisor in the resolved bundle. On a resolved fibre, $[\theta_j]$ generates the degree-two relative cohomology and $-\frac{1}{2}[\theta_j]^2$ generates the degree-four relative cohomology, because the exceptional curve has self-intersection -2 . The local coefficient systems are trivial. Indeed, the degree-four system is the orientation system of the oriented normal cone, while in (76) the deck transformation acts holomorphically on $\mathbb{CP}^1$ and therefore fixes both $[\theta_j]$ and $[\theta_j]^2$.

Relative Leray–Hirsch, followed by excision, now gives

$$H^k(X, U_0) \cong \bigoplus_j H^{k-4}(N_j),$$

$$H^k(M, U_0) \cong \bigoplus_j \left(H^{k-2}(N_j)[\theta_j] \oplus H^{k-4}(N_j)[\theta_j]^2 \right).$$

The fibrewise blowdown has oriented degree one. Therefore the relative pullback of T_j has fibre integral one, so its $-\frac{1}{2}[\theta_j]^2$ -coordinate is one. Consequently relative pullback is injective, its image is a graph over the degree-four summand, and its cokernel is identified, using the chosen Leray–Hirsch generators, with $\bigoplus_j H^{k-2}(N_j)[\theta_j]$. Choosing this graph as the common summand gives the relative splitting used below. We define

$$\Psi_k(\xi, (\lambda_j)_j) = r^*\xi + \sum_j [\text{pr}_j^* \lambda_j \wedge \theta_j]. \quad (92)$$

Here ξ and the λ_j are cohomology classes. To realize the displayed map by differential forms, we choose a representative of ξ that is constant in the normal directions near every N_j before taking its pullback, as in [11, Section 3], and we choose closed representatives of the λ_j on the N_j .

We now give the diagram chase that proves that Ψ_k is an isomorphism. The vertical map on the common exterior U_0 is the identity. We fix $y \in H^k(M)$ and write $u = y|_{U_0}$. The boundary of u in the long exact sequence of (M, U_0) is zero. Naturality and injectivity of relative pullback on the common $\bigoplus_j H^{k+1-4}(N_j)$ summand imply that the boundary of u in the long exact sequence of (X, U_0) is also zero. Exactness therefore gives a class $\xi \in H^k(X)$ satisfying $\xi|_{U_0} = u$. The class $y - r^*\xi$ restricts to zero on U_0 and hence is the image of a relative class in $H^k(M, U_0)$. Decompose that relative class into its common degree-four part and its additional degree-two part. The common part is the pullback of a relative class for (X, U_0) and can be absorbed into ξ ; the additional part maps to a sum of the supported classes in (92). This proves that Ψ_k is surjective.

The new summands are independent of one another and of the pullback summand. Indeed, if $i_j : E_j \hookrightarrow M$ and $p_j : E_j \rightarrow N_j$, then

$$-\frac{1}{2}(p_j)_* i_j^* (\text{pr}_k^* \lambda_k \wedge \theta_k) = \delta_{jk} \lambda_k, \quad (p_j)_* i_j^* r^* \xi = 0. \quad (93)$$

Indeed, $i_j^* \theta_j = e(\mathcal{O}(-2))$ integrates to -2 on the $\mathbb{CP}^1$ fibre. Thus the homomorphism $C_j = -\frac{1}{2}(p_j)_* i_j^*$ extracts the j th exceptional coefficient.

It remains to prove that $r^* : H^k(X) \rightarrow H^k(M)$ is injective. Suppose that $r^*\xi = 0$. Its restriction to U_0 is zero, so exactness represents ξ by a relative class $a \in H^k(X, U_0)$. Naturality shows that the relative pullback of a maps to zero in $H^k(M)$. Exactness for (M, U_0) therefore expresses that pullback as the boundary of a class in $H^{k-1}(U_0)$. The relative pullback is injective on the common relative summand, so naturality implies that a itself is the boundary of the same class in the sequence for (X, U_0) . Hence $\xi = 0$. The maps C_j and the injectivity of r^* prove that Ψ_k is injective. Therefore every Ψ_k is an isomorphism, proving (90) with the stated normalization. Supports for different resolution neighborhoods are disjoint, so $x_j x_k = 0$ for $j \neq k$. On one fibre, the normal line of the exceptional $\mathbb{CP}^1$ is $\mathcal{O}(-2)$; its Euler class evaluates to -2 . The Thom self-intersection formula therefore gives $x_j^2 = -2 \text{PD}[N_j]$, including the displayed sign. This also proves the supported pushforward rule (109) used below. Here $\text{PD}[N_j]$ means the pullback of the Poincaré dual in X , regarded as an element of the $H^4(X)$ summand in (90). We denote the exceptional projection by

$$\pi_E : H^5(M) \longrightarrow \bigoplus_{j=1}^{10} H^3(N_j) x_j. \quad (94)$$

The two products needed to define the Massey product are

$$ab = (x_1 + x_2)(x_7 + x_3) = 0$$

because the four exceptional neighborhoods are pairwise disjoint, and

$$bc = x_7^2 - x_3^2 = -2 \text{PD}[N_7] + 2 \text{PD}[N_3] = 0$$

because $\text{PD}[N_3] = \text{PD}[N_7]$. Thus $\langle a, b, c \rangle$ is defined. We define $v_j = [\varphi|_{N_j}]x_j$. We recall the geometric construction of the primitive used by Martín–Merchán. On the mapping-torus cover, N_3^1 and N_7^1 are joined by the embedded oriented cobordism

$$\mathcal{C} = \{(t, x_1, x_2, x_3 + \mathbf{i}f(t)) : \tfrac{1}{2} \leq t \leq \tfrac{3}{2}, x_1, x_2, x_3 \in \mathbb{R}/\mathbb{Z}\},$$

where f is nondecreasing, $f = 0$ for $t \leq 9/8$, and $f = 1/4$ for $t \geq 5/4$. Two points in the displayed parameter domain have the same image only if their t -coordinates differ by an integer. Since the interior t -coordinates lie in an interval of length one, equality in the interior forces equality of all parameters. At the endpoints, the only additional identification is the defining mapping-torus identification. It identifies the upper boundary with the specified lift of N_7 and does not identify distinct interior points. Hence the parametrization is an embedding, and the boundary-orientation convention gives $\partial\mathcal{C} = N_3^1 - N_7^1$. Its normal bundle is trivialized by

$$\partial_{y_1}, \quad \partial_{y_2}, \quad \partial_{y_3} - f'(t)\partial_t.$$

We choose $\zeta : \mathbb{R}^3/\mathbb{Z}^3 \rightarrow \mathbb{R}$ with integral one and support in $B_{2\delta}(0) \setminus B_\delta(0)$. We also choose h to be zero near $t = 1/2, 3/2$, equal to one on $[1/2 + 2\epsilon, 3/2 - 2\epsilon]$. In the displayed normal coordinates, we define

$$\beta_{\mathcal{C}} = h(t)\zeta(y_1, y_2, y_3 - f(t)) dy_1 \wedge dy_2 \wedge (dy_3 - f'(t) dt). \quad (95)$$

This is the Thom form of $\nu\mathcal{C}$ away from the boundary collars. We choose h to be increasing on the lower boundary collar and decreasing on the upper boundary collar. We denote the four-form supported in the lower collar by v_3^1 and the negative of the four-form supported in the upper collar by v_7^1 . Then

$$d\beta_{\mathcal{C}} = h'(t)\zeta(y_1, y_2, y_3 - f(t)) dt \wedge dy_1 \wedge dy_2 \wedge dy_3 = v_3^1 - v_7^1.$$

The normalization $\int \zeta = 1$, together with $\int h'(t) dt = 1$ on the lower collar and $\int h'(t) dt = -1$ on the upper collar, shows that both v_3^1 and v_7^1 have fibre integral +1 with respect to the oriented normal fibres of their respective boundary components.

We define $K_0 = \langle \iota_1, \iota_2, \kappa \rangle$, which has order eight, and set

$$\beta_0 = \frac{1}{4} \sum_{\gamma \in K_0} \gamma^* \beta_{\mathcal{C}}, \quad v_j^0 = \frac{1}{4} \sum_{\gamma \in K_0} \gamma^* v_j^1 \quad (j = 3, 7).$$

These forms are K_0 -invariant and satisfy $d\beta_0 = v_3^0 - v_7^0$. The subgroup of K_0 preserving a chosen lifted component has order four. Thus, on a lifted normal fibre, the four coincident stabilizer terms cancel the factor 1/4, and v_j^0 has fibre integral one. The invariant forms descend through the old quotients. On a normal fibre of N_j , the remaining quotient is the degree-two map $\mathbb{C}^2 \rightarrow \mathbb{C}^2/\{\pm 1\}$. Consequently the descended closed forms v_j on X have the precise normalization

$$2[v_j] = \text{Th}[N_j] \in H_c^4(V_j), \quad j = 3, 7. \quad (96)$$

We denote the three-form on X descended from β_0 by $\bar{\beta}_0$. It satisfies $d\bar{\beta}_0 = v_3 - v_7$. At $t = 1$ one has $h = 1$ and $f = 0$. Substitution of the component parametrizations gives

$$\int_{N_1^k} \beta_{\mathcal{C}} = -1, \quad \int_{N_2^k} \beta_{\mathcal{C}} = +1, \quad \int_{N_j^k} \varphi = 2 \quad (j = 1, 2).$$

Set $\sigma_1 = -1$ and $\sigma_2 = 1$. Every element of K_0 preserves φ , preserves the Cayley orientations of these components, and permutes the two lifts of each N_j . Hence, for every $\gamma \in K_0$ and $j, k \in \{1, 2\}$,

$$[(\gamma^* \beta_{\mathcal{C}})|_{N_j^k}] = \frac{\sigma_j}{2} [\varphi|_{N_j^k}] \in H^3(N_j^k).$$

There are eight terms in the definition of β_0 and its coefficient is 1/4. It follows that

$$[\beta_0|_{N_j^k}] = \sigma_j [\varphi|_{N_j^k}].$$

Both sides are pullbacks of classes on the quotient component N_j . Pullback along the finite covering of the component is injective over $\mathbb{R}$, so

$$[\bar{\beta}_0|_{N_1}] = -[\varphi|_{N_1}], \quad [\bar{\beta}_0|_{N_2}] = [\varphi|_{N_2}]. \quad (97)$$

After shrinking the already chosen neighborhoods of N_3 and N_7 , we choose invariant tubular neighborhoods V_1, V_2 so that V_1, V_2, V_3, V_7 are pairwise disjoint. On V_j , for $j = 1, 2$, the closed form $\bar{\beta}_0 - \sigma_j \pi_j^*(\varphi|_{N_j})$ restricts trivially to the zero section. Since V_j deformation retracts onto N_j , there is a two-form q_j on V_j such that

$$dq_j = \bar{\beta}_0 - \sigma_j \pi_j^*(\varphi|_{N_j}).$$

We average q_j over the finite stabilizer of V_j and choose an invariant cutoff χ_j supported in V_j and equal to one on a smaller tubular neighborhood. We define

$$\alpha_{\mathcal{C}} = \bar{\beta}_0 - d(\chi_1 q_1 + \chi_2 q_2).$$

It follows directly that

$$d\alpha_{\mathcal{C}} = v_3 - v_7, \quad \alpha_{\mathcal{C}}|_{V_1} = -\pi_1^*(\varphi|_{N_1}), \quad \alpha_{\mathcal{C}}|_{V_2} = \pi_2^*(\varphi|_{N_2}). \quad (98)$$

The form $\bar{\beta}_0$ is zero in a neighborhood of the $t = 1/2$ slice, whereas the two cutoff corrections are supported near N_1 and N_2 in the $t = 1$ slice. Therefore $\alpha_{\mathcal{C}}$ is zero near the $t = 1/2$ slice. This slice contains every N_j with $3 \leq j \leq 10$, so $\alpha_{\mathcal{C}}$ vanishes near all of those components, not only near $N_3 \cup N_7$. The two signs are the oriented intersection numbers $\mathcal{C} \cdot N_1 = -1$ and $\mathcal{C} \cdot N_2 = +1$; they follow by comparing $(-\partial_t, \partial_{x_1}, \partial_{x_2}, \partial_{x_3})$ on $T\mathcal{C}$ with the Cayley orientations of N_1, N_2 .

We choose the radial exceptional Thom representatives used in [11]:

$$\theta_j^{\text{MM}} = \frac{1}{\pi i} d(b(r^2) \partial \log r^2), \quad (99)$$

where r is the radial coordinate on $\mathbb{C}^2$ in the resolution model $Y \rightarrow \mathbb{C}^2/\{\pm 1\}$, and ∂ is the Dolbeault $(1,0)$ operator in those complex coordinates. The function $b : [0, \infty) \rightarrow \mathbb{R}$ is smooth and nondecreasing, equals -1 for $s \leq \epsilon^2$, and equals 0 for $s \geq 4\epsilon^2$. The form θ_j^{MM} integrates to one on the normal fibre $\mathbb{C}/\{\pm 1\}$ of the exceptional divisor. Near that divisor it is the pullback of a two-form on $\mathbb{CP}^1$, so $(\theta_j^{\text{MM}})^2$ vanishes on a neighborhood of the divisor. On the complement of the divisor, we transport the square through the blowdown diffeomorphism. Since it vanishes near the divisor, the transported form extends by zero across N_j and defines the smooth compactly supported form $r_*((\theta_j^{\text{MM}})^2)$ on X . We choose the radial cutoff in (99) sufficiently small that this support is contained in the already fixed V_j ; the form v_j is supported there as well. Fibre integration and the self-intersection number of $\mathcal{O}_{\mathbb{CP}^1}(-2)$ give

$$[r_*((\theta_j^{\text{MM}})^2)] = -2 \text{Th}[N_j] \quad \text{in } H_c^4(V_j). \quad (100)$$

Equation (96) gives the exact compactly supported cohomology calculation in $H_c^4(V_j)$

$$[r_*((\theta_j^{\text{MM}})^2) + 4v_j] = -2 \text{Th}[N_j] + 4 \cdot \frac{1}{2} \text{Th}[N_j] = 0.$$

By compactly supported de Rham theory on the orbifold chart, there are three-forms $\mu_j \in \Omega_c^3(V_j)$, extended by zero to X , such that

$$d\mu_j = r_*((\theta_j^{\text{MM}})^2) + 4v_j, \quad j = 3, 7.$$

The transported square and the annular representative v_j both vanish on a sufficiently small neighborhood of N_j . Thus $d\mu_j = 0$ there. We denote the radial tubular homotopy operator by H_j . Its homotopy identity reduces on this neighborhood to

$$\mu_j - \pi_j^*(\mu_j|_{N_j}) = dH_j\mu_j.$$

We choose a cutoff χ_j^{cut} that is supported in this neighborhood and equals one on a smaller neighborhood. Replacing μ_j by

$$\mu_j - d(\chi_j^{\text{cut}} H_j \mu_j)$$

does not change $d\mu_j$ and gives $\mu_j = \pi_j^*(\mu_j|_{N_j})$ on the smaller neighborhood. We retain the notation μ_j for the modified form. The cutoff correction is supported in V_j , so the modified form still belongs to $\Omega_c^3(V_j)$. Its normal-pullback form implies that $r^*\mu_j$ extends smoothly across the exceptional divisor. Likewise $r^*\alpha_{\mathcal{C}}$ is smooth: it is zero near every N_j , $3 \leq j \leq 10$, and is the pullback of a form on the centre near N_1, N_2 . Finally,

$$r^*r_*((\theta_j^{\text{MM}})^2) = (\theta_j^{\text{MM}})^2, \quad (101)$$

because both sides vanish near E_j and r is a diffeomorphism on their remaining support. Hence

$$\eta_{\text{MM}} = r^*(\mu_7 + 4\alpha_{\mathcal{C}} - \mu_3) \quad (102)$$

is smooth. The supports of μ_3, μ_7 are disjoint from the first two resolution neighborhoods because the four tubular neighborhoods were chosen pairwise disjoint. Thus (98) gives

$$\eta_{\text{MM}}|_{\text{supp } \theta_1^{\text{MM}}} = -4 \text{pr}_1^*(\varphi|_{N_1}), \quad \eta_{\text{MM}}|_{\text{supp } \theta_2^{\text{MM}}} = 4 \text{pr}_2^*(\varphi|_{N_2}).$$

The defining equations for μ_3, μ_7 , and α_C give

$$\begin{aligned} d\eta_{\text{MM}} &= r^*(d\mu_7 + 4d\alpha_C - d\mu_3) \\ &= r^*(r_*((\theta_7^{\text{MM}})^2) + 4v_7 + 4(v_3 - v_7) - r_*((\theta_3^{\text{MM}})^2) - 4v_3) \\ &= (\theta_7^{\text{MM}})^2 - (\theta_3^{\text{MM}})^2. \end{aligned}$$

The last equality follows from (101). The restrictions above and the supported Thom product formula then give

$$[\eta_{\text{MM}}\theta_1^{\text{MM}}] = -4v_1, \quad [\eta_{\text{MM}}\theta_2^{\text{MM}}] = 4v_2. \quad (103)$$

Thus the explicitly computed class is

$$Y_{\text{MM}} = [\eta_{\text{MM}}(\theta_1^{\text{MM}} + \theta_2^{\text{MM}})] = -4v_1 + 4v_2. \quad (104)$$

With the convention (86), the corresponding defining-system representative is $-Y_{\text{MM}}$; this sign does not affect membership in the vector subspace I_M . The projection of the *entire* degree-five indeterminacy $aH^3(M) + H^3(M)c$ to the exceptional summands is contained in

$$\text{span}\{v_1 + v_2, v_7 - v_3\}. \quad (105)$$

Indeed, if $\xi \in H^3(X)$, then $\xi x_j = (\xi|_{N_j})x_j$. We now perform the restriction calculation explicitly. Every such class has a representative

$$\xi = dt \wedge \beta_2 + \lambda_1(dz_{123} + d\bar{z}_{123}) + \lambda_2(dz_1 \wedge dz_2 \wedge d\bar{z}_3 + d\bar{z}_1 \wedge d\bar{z}_2 \wedge dz_3), \quad (106)$$

where β_2 is a constant closed two-form on the flat cover satisfying $F^*\beta_2 = \beta_2$ and $\kappa^*\beta_2 = -\beta_2$. The dt term restricts to zero. Substitution of the component parametrizations from (73) gives, on their chosen twofold covers,

$$\begin{aligned} \xi|_{N_1^1} &= 4(\lambda_1 - \lambda_2) dx_1 \wedge dx_2 \wedge dy_3, & \varphi|_{N_1^1} &= 2 dx_1 \wedge dx_2 \wedge dy_3, \\ \xi|_{N_2^1} &= -4(\lambda_1 - \lambda_2) dx_1 \wedge dx_2 \wedge dy_3, & \varphi|_{N_2^1} &= -2 dx_1 \wedge dx_2 \wedge dy_3, \\ \xi|_{N_3^1} &= 2(\lambda_1 + \lambda_2) dx_1 \wedge dx_2 \wedge dx_3, & \varphi|_{N_3^1} &= dx_1 \wedge dx_2 \wedge dx_3, \\ \xi|_{N_7^1} &= -2(\lambda_1 + \lambda_2) dy_1 \wedge dy_2 \wedge dx_3, & \varphi|_{N_7^1} &= -dy_1 \wedge dy_2 \wedge dx_3. \end{aligned}$$

Passing through the twofold covers changes both integrals by the same factor. Since $H^3(N_j) = \mathbb{R}[\varphi|_{N_j}]$, the scalars are therefore

$$\lambda_{12}(\xi) = 2(\lambda_1 - \lambda_2), \quad \lambda_{37}(\xi) = 2(\lambda_1 + \lambda_2), \quad (107)$$

and

$$\begin{aligned} [\xi|_{N_1}] &= \lambda_{12}(\xi)[\varphi|_{N_1}], & [\xi|_{N_2}] &= \lambda_{12}(\xi)[\varphi|_{N_2}], \\ [\xi|_{N_3}] &= \lambda_{37}(\xi)[\varphi|_{N_3}], & [\xi|_{N_7}] &= \lambda_{37}(\xi)[\varphi|_{N_7}]. \end{aligned}$$

Consequently

$$\pi_E(a\xi) = \lambda_{12}(\xi)(v_1 + v_2), \quad \pi_E(\xi c) = \lambda_{37}(\xi)(v_7 - v_3). \quad (108)$$

For the remaining terms, decompose an arbitrary class as $h = \xi + \sum_j \lambda_j x_j \in H^3(M)$, with $\xi \in H^3(X)$ and $\lambda_j \in H^1(N_j)$. The supported representative product rule of [11, Section 3] is

$$(\lambda x_j)x_j = -2(i_j)_! \lambda \in H^5(X), \quad i_j : N_j \hookrightarrow X, \quad (109)$$

while $(\lambda x_j)x_k = 0$ for $j \neq k$. Hence these products have zero exceptional projection. More explicitly, multiplication by $a = x_1 + x_2$ can involve only the exceptional terms with $j = 1, 2$, and multiplication by $c = x_7 - x_3$ can involve only those with $j = 3, 7$. Formula (109) places every resulting square in the base summand $H^5(X)$, so none contributes to π_E . Together with (108), this proves (105). Finally, if

$$-4v_1 + 4v_2 = A(v_1 + v_2) + B(v_7 - v_3),$$

the independent v_1 and v_2 coordinates in (90) would give simultaneously $A = -4$ and $A = 4$, which is impossible. Therefore $Y_{\text{MM}} \notin I_M$, and likewise $-Y_{\text{MM}} \notin I_M$. Hence

$$0 \notin \langle a, b, c \rangle.$$

□

Lemma 4.1 gives

$$\tilde{\tau}^* a = -a, \quad \tilde{\tau}^* b = b, \quad \tilde{\tau}^* c = c. \quad (110)$$

We choose pairwise disjoint tubular neighborhoods of the exceptional divisors that are invariant as a collection under $\tilde{\tau}$. We choose a normalized Thom form θ_2 representing x_2 and define $\theta_1 = -\tilde{\tau}^* \theta_2$; equation (71) shows that θ_1 represents x_1 . For each $j \in \{3, 7\}$, we average a normalized Thom form with its pullback under $\tilde{\tau}$. The resulting form, still denoted by θ_j , is normalized and represents x_j . These forms satisfy

$$\tilde{\tau}^* \theta_1 = -\theta_2, \quad \tilde{\tau}^* \theta_2 = -\theta_1, \quad \tilde{\tau}^* \theta_j = \theta_j \quad (j = 3, 7). \quad (111)$$

We define

$$\alpha = \theta_1 + \theta_2, \quad \beta = \theta_7 + \theta_3, \quad \gamma = \theta_7 - \theta_3. \quad (112)$$

The disjoint supports give $\alpha\beta = 0$, while $\beta\gamma = \theta_7^2 - \theta_3^2$ is exact by $\text{PD}[N_7] = \text{PD}[N_3]$. We choose a three-form η_0 satisfying $d\eta_0 = \beta\gamma$ and replace it by

$$\eta = \frac{1}{2}(\eta_0 + \tilde{\tau}^* \eta_0). \quad (113)$$

Since $\beta\gamma$ is invariant, $d\eta = \beta\gamma$ and $\tilde{\tau}^* \eta = \eta$. Moreover $d(\alpha\eta) = \alpha\beta\gamma = 0$. Under the convention (86), the defining system $(\xi_{12}, \xi_{23}) = (0, \eta)$ has representative $-\alpha\eta$. Replacing the normalized Thom representatives or replacing a primitive by another one changes this class by $aH^3(M) + H^3(M)c = I_M$, exactly as in (87). Thus the averaged system and the explicit system (103) define representatives in the same affine Massey coset. Equivalently, their cohomology classes differ by an element of I_M . Proposition 5.1 therefore gives $-\alpha\eta \notin I_M$. Since I_M is a vector subspace, we define

$$Y = [\alpha\eta] \in H^5(M). \quad (114)$$

Then $Y \notin I_M$, and $\tilde{\tau}^* Y = [(-\alpha)\eta] = -Y$.

5.1 The quotient CDGA and its indeterminacy

Lemma 5.2. *Suppose that a finite group Γ acts smoothly on a compact manifold P , and denote the quotient map by $q : P \rightarrow P/\Gamma$. With real coefficients, pullback is a quasi-isomorphism*

$$q^* : A_{\text{PL}}(P/\Gamma; \mathbb{R}) \longrightarrow A_{\text{PL}}(P; \mathbb{R})^\Gamma. \quad (115)$$

Proof. By the equivariant triangulation theorem for smooth finite-group actions [8], P admits a finite Γ -CW structure. The image in P/Γ of an equivariant cell $\Gamma/H \times D^n$ is the ordinary cell D^n , whereas the corresponding piece of the Borel construction is $BH \times D^n$. The map from the Borel construction to the orbit space restricts over this cell to the projection $BH \times D^n \rightarrow D^n$. Since H is finite, $H^q(BH; \mathbb{R}) = 0$ for $q > 0$ and $H^0(BH; \mathbb{R}) = \mathbb{R}$. Induction over the relative cohomology sequences of the cellular skeleta shows that $P_{h\Gamma} \rightarrow P/\Gamma$ is a real cohomology equivalence. Equivalently, the Leray spectral sequence has higher stalk at the orbit of p equal to $H^q(B\Gamma_p; \mathbb{R}) = 0$ for every $q > 0$.

Let $\pi_h : P_{h\Gamma} \rightarrow P/\Gamma$ be the preceding map. Fix $e \in E\Gamma$ and define $j_e : P \rightarrow P_{h\Gamma}$ by $j_e(p) = [e, p]$. Then $\pi_h \circ j_e = q$. The Borel spectral sequence has

$$E_2^{r,s} = H^r(\Gamma; H^s(P; \mathbb{R})).$$

Higher finite-group cohomology vanishes in characteristic zero, so it collapses, and its edge homomorphism j_e^* is an isomorphism from $H^*(P_{h\Gamma}; \mathbb{R})$ onto $H^*(P; \mathbb{R})^\Gamma$. Since π_h^* is an isomorphism by the preceding skeletal argument and $q^* = j_e^* \pi_h^*$, pullback gives

$$q^* : H^*(P/\Gamma; \mathbb{R}) \xrightarrow{\cong} H^*(P; \mathbb{R})^\Gamma.$$

The Reynolds operator makes invariants exact, whence

$$H^*(A_{\text{PL}}(P; \mathbb{R})^\Gamma) = H^*(A_{\text{PL}}(P; \mathbb{R}))^\Gamma.$$

This proves (115). The simplicial de Rham theorem [2, 6], applied to the smooth singular simplicial set and followed by exact finite-group invariants, identifies this model with $\Omega(P)^\Gamma$ through a zigzag of CDGA quasi-isomorphisms.

Finally, if $f : A \rightarrow B$ is a quasi-isomorphism of CDGAs, then for every defined triple product

$$H(f)(\langle a_1, a_2, a_3 \rangle_A) = \langle H(f)a_1, H(f)a_2, H(f)a_3 \rangle_B.$$

Indeed, f carries one defining system to a defining system. Both sides are nonempty affine cosets, and $H(f)$ maps the entire indeterminacy isomorphically by (87); hence the initial inclusion is equality. Thus the two full Massey sets correspond under $H(f)$. $\square$

We denote the quotient map by $q : W = S^1 \times M \rightarrow Z_0$. The preceding lemma and the equivariant simplicial de Rham comparison give the explicit zigzag

$$A_{\text{PL}}(Z_0; \mathbb{R}) \xrightarrow{q^*} A_{\text{PL}}(W; \mathbb{R})^{(\sigma)} \xrightarrow{\simeq} A_{\text{dR}}(\text{Sing}^\infty W)^{(\sigma)} \xleftarrow{\simeq} \Omega(W)^{(\sigma)}. \quad (116)$$

Here $A_{\text{dR}}(\text{Sing}^\infty W)$ is the CDGA of compatible smooth forms on smooth singular simplices. The left comparison includes polynomial simplex forms and the right comparison pulls a differential form back to every smooth simplex. Both are natural quasi-isomorphisms; taking invariants preserves this because the Reynolds operator is exact. Thus differential forms below live in the right-hand invariant de Rham model; their classes and their entire Massey cosets are transported to $A_{\text{PL}}(Z_0; \mathbb{R})$

through (116). No de Rham complex on the singular quotient is being assumed. More precisely, when an invariant differential-form defining system is used below, equality of Massey sets under the quasi-isomorphisms in the zigzag places the class of its representative in the corresponding Massey set in $A_{\text{PL}}(Z_0; \mathbb{R})$. By the definition of that set, we then choose an actual $A_{\text{PL}}(Z_0; \mathbb{R})$ defining system realizing the transported class.

We write $u = [ds] \in H^1(S^1)$. This class does not itself descend. If $\xi \in H^*(M)^-$, we write $u\xi$ for the unique quotient class whose pullback to $S^1 \times M$ is $[ds] \otimes \xi$. The invariant ring decomposition is

$$H^k(Z_0) = H^k(M)^+ \oplus uH^{k-1}(M)^-. \quad (117)$$

Here the superscripts denote the ± 1 eigenspaces of $\tilde{\tau}^*$. Multiplication is inherited from $H^*(S^1 \times M)$, in particular $u^2 = 0$. The invariant forms on W

$$A = ds \wedge \alpha, \quad B = \beta, \quad C = \gamma \quad (118)$$

have degrees $(3, 2, 2)$, with $AB = 0$ and $BC = d\eta$. Thus the triple product $\langle ua, b, c \rangle$ is defined. The defining system $(\xi_{12}, \xi_{23}) = (0, \eta)$ has representative

$$0 \cdot C - (-1)^3 A\eta = ds \wedge \alpha \wedge \eta,$$

whose cohomology class corresponds to $uY \in H^6(Z_0)$ under (116). Consequently there are closed elements $A_0, B_0, C_0 \in A_{\text{PL}}(Z_0; \mathbb{R})$ representing ua, b, c , respectively, and elements ξ_{12}^0, ξ_{23}^0 such that

$$d\xi_{12}^0 = A_0 B_0, \quad d\xi_{23}^0 = B_0 C_0, \quad (119)$$

$$[\xi_{12}^0 C_0 - (-1)^3 A_0 \xi_{23}^0] = uY \in H^6(Z_0; \mathbb{R}). \quad (120)$$

We fix this A_{PL} defining system for the remainder of the proof.

Proposition 5.3. *The triple Massey product $\langle ua, b, c \rangle$ is defined, and*

$$0 \notin \langle ua, b, c \rangle. \quad (121)$$

Proof. In the two degrees entering the indeterminacy,

$$H^3(Z_0) = H^3(M)^+ \oplus uH^2(M)^-, \quad H^4(Z_0) = H^4(M)^+ \oplus uH^3(M)^-.$$

Since $u^2 = 0$, the two summands multiply as

$$(ua)H^3(Z_0) = u a H^3(M)^+, \quad H^4(Z_0)c = H^4(M)^+ c \oplus uH^3(M)^- c.$$

The full indeterminacy is

$$I_0 = (ua)H^3(Z_0) + H^4(Z_0)c \quad (122)$$

$$= H^4(M)^+ c + u(aH^3(M)^+ + H^3(M)^- c). \quad (123)$$

If $uY \in I_0$, projection to the u -summand would give $Y \in aH^3(M)^+ + H^3(M)^- c \subset I_M$, a contradiction. Thus the affine coset $uY + I_0$ does not contain zero. $\square$

5.2 Passage through the resolution

Lemma 5.4. *The space Z_0 is a compact oriented real homology eight-manifold, and the degree-one map $\rho : \widehat{Z} \rightarrow Z_0$ has homomorphisms*

$$\rho_! : H^k(\widehat{Z}; \mathbb{R}) \longrightarrow H^k(Z_0; \mathbb{R}) \quad (0 \leq k \leq 8)$$

such that, for every $v \in H^p(Z_0; \mathbb{R})$ and $w \in H^q(\widehat{Z}; \mathbb{R})$,

$$\rho_! \rho^* v = v, \quad \rho_!(\rho^* v \smile w) = v \smile \rho_!(w). \quad (124)$$

Proof. On the dense regular set $\tilde{\tau}$ agrees with τ , whose linear part has determinant -1 , so $\tilde{\tau}$ reverses orientation. Reflection reverses the orientation of S^1 ; hence σ preserves the orientation of W .

We fix $z \in Z_0$ and denote the stabilizer of one of its lifts by $H \leq \langle \sigma \rangle$. The exponential map of an H -invariant metric gives the usual Bochner linearization, so a small link of z is S^7/H , with H acting linearly through $SO(8)$. If H is nontrivial, the derivative of its generator is $+I$ on the tangent four-plane to the fixed component and $-I$ on its normal four-plane; in particular its determinant and its degree on S^7 are $+1$. We choose an H -equivariant triangulation [8] and barycentrically subdivide it so that every simplex stabilizer fixes that simplex pointwise. The cellular chain complex of the quotient is the coinvariant complex. Over $\mathbb{R}$, the coinvariant functor is exact, and the averaging operator identifies coinvariants with invariants. Therefore

$$H_*(S^7/H; \mathbb{R}) \cong H_*(S^7; \mathbb{R})_H \cong H_*(S^7; \mathbb{R})^H.$$

Every element of H has degree $+1$ on S^7 , so this vector space is $\mathbb{R}$ in degrees $0, 7$ and zero otherwise. Therefore

$$H_i(Z_0, Z_0 \setminus \{z\}; \mathbb{R}) \cong \tilde{H}_{i-1}(S^7/H; \mathbb{R}) = \begin{cases} \mathbb{R}, & i = 8, \\ 0, & i \neq 8. \end{cases} \quad (125)$$

The local top generators agree with the quotient orientation. Thus Z_0 is a compact oriented real homology eight-manifold. Finite smooth quotients are triangulable and locally contractible, so Poincaré duality for compact oriented ENR homology manifolds [3, Chapter V] applies: Z_0 has a fundamental class $[Z_0]$ and cap product induces isomorphisms

$$D_{Z_0} : H^k(Z_0) \longrightarrow H_{8-k}(Z_0), \quad D_{Z_0}(v) = v \frown [Z_0]. \quad (126)$$

We orient $\widehat{Z}$ by its Cayley orientation. We choose a point in the nonempty common flat open set lying outside every first- and second-stage resolution neighborhood. At that point ρ is the identity under the staged/Joyce atlas identification. The quotient orientation of Z_0 there is the orientation induced by the flat Cayley form Φ_0 ; the Joyce preglued positive form and its sufficiently small torsion-free perturbation lie in the same positive Cayley cone and induce that same orientation. This regular point therefore has exactly one preimage and local degree $+1$. Hence $\rho_*[\widehat{Z}] = [Z_0]$. We define

$$\rho_! = D_{Z_0}^{-1} \rho_* D_{\widehat{Z}}. \quad (127)$$

Cap-product naturality gives the projection formula $\rho_!(\rho^* v \smile w) = v \smile \rho_!(w)$. The fundamental-class identity gives $\rho_!(1) = 1$; therefore

$$\rho_! \rho^* v = \rho_!(\rho^* v \smile 1) = v \smile \rho_!(1) = v,$$

which proves both identities in (124). □

Theorem 5.5. *The triple Massey product $\langle \rho^*(ua), \rho^*b, \rho^*c \rangle$ is defined, and*

$$0 \notin \langle \rho^*(ua), \rho^*b, \rho^*c \rangle. \quad (128)$$

Thus $\widehat{Z}$ is nonformal.

Proof. The continuous map ρ induces the strict CDGA morphism

$$\rho^* : A_{\text{PL}}(Z_0; \mathbb{R}) \longrightarrow A_{\text{PL}}(\widehat{Z}; \mathbb{R}).$$

Pulling back the defining system fixed in (119)–(120) proves that the triple product is defined and gives a representative with cohomology class $\rho^*(uY)$. The ordinary de Rham comparison for the smooth manifold $\widehat{Z}$ carries this entire Massey set to the corresponding one in $\Omega(\widehat{Z})$. Its full indeterminacy is

$$I_{\widehat{Z}} = \rho^*(ua)H^3(\widehat{Z}) + H^4(\widehat{Z})\rho^*c. \quad (129)$$

Suppose $\rho^*(uY) = \rho^*(ua)r + q\rho^*c$ for $r \in H^3(\widehat{Z})$ and $q \in H^4(\widehat{Z})$. The projection formula and $\rho_!\rho^* = \text{id}$ give

$$uY = (ua)\rho_!(r) + \rho_!(q)c.$$

For the second term, explicitly, $\rho_!(q\rho^*c) = \rho_!(\rho^*cq) = c\rho_!(q) = \rho_!(q)c$; no sign occurs because $|q||c| = 8$. The displayed class lies in I_0 , contrary to Proposition 5.3. Hence $\rho^*(uY) \notin I_{\widehat{Z}}$, so the full affine Massey coset does not contain zero. Every defined Massey product on a formal CDGA contains zero, so (128) obstructs formality of $\widehat{Z}$. $\square$

6 Simple connectivity and exact holonomy

Proposition 6.1. *The manifold $\widehat{Z}$ is simply connected.*

Proof. The manifold M is simply connected [11, Section 2.2]; hence $\mathbb{R} \times M$ is simply connected. We define an action on $\mathbb{R} \times M$ of the infinite dihedral group

$$D_\infty = \langle S, R \mid R^2 = 1, RSR = S^{-1} \rangle, \quad (130)$$

by the formulas $S(s, p) = (s + 1, p)$ and $R(s, p) = (-s, \widetilde{\tau}p)$. These formulas satisfy the displayed relations and therefore define the asserted action. The fixed set of $\widetilde{\tau}$ is nonempty. Indeed, on $\mathbb{T}^7$ we impose $t = \frac{1}{2}$ and $z_3 = \frac{1}{8}$. The resulting τ -fixed slice is a four-torus. The lift of the old singular set is a finite union of three-tori, so it cannot cover the slice. We choose $(z_1, z_2) \in E^2$ in its complement. The resulting point is fixed by τ . Its image $x \in X$ is regular; because r is a diffeomorphism over the regular set, the unique point of $r^{-1}(x)$ is fixed by $\widetilde{\tau}$. Consequently R fixes $(0, p)$ and SR fixes $(\frac{1}{2}, p)$ in $\mathbb{R} \times M$. The quotient by $\langle S \rangle$ is $S^1 \times M$, and the residual action of R is precisely σ ; hence $(\mathbb{R} \times M)/D_\infty \cong Z_0$. The space $\mathbb{R} \times M$ is connected, simply connected, locally compact, and metrizable. For every compact subset $C \subset \mathbb{R} \times M$, its projection to the s -coordinate is contained in a bounded interval. Consequently only finitely many integers n satisfy $S^n C \cap C \neq \emptyset$, and only finitely many integers n satisfy $S^n R C \cap C \neq \emptyset$. No nonidentity translation S^n has a fixed point. A reflection $S^n R$ can fix (s, p) only if $s = n/2$ and $\widetilde{\tau}(p) = p$. If two reflections $S^n R$ and $S^m R$ fixed the same point, then their product S^{n-m} would fix that point, and hence $n = m$. Thus every stabilizer is either trivial or the order-two group generated by a unique reflection.

We also verify Armstrong's local discontinuity condition. Given a point, choose a relatively compact neighborhood. The two compact-intersection statements above leave only finitely many group elements whose translates can meet its closure. For each of those finitely many elements that

does not stabilize the point, shrink the neighborhood so that it is disjoint from its translate. The resulting neighborhood can meet a translate of itself only under an element of its finite stabilizer. Thus the action is discontinuous in the precise sense required by Armstrong's theorem [1], which gives

$$\pi_1(Z_0) = D_\infty/N,$$

where N is normally generated by elements having fixed points. Thus $R, SR \in N$, and $S = (SR)R \in N$, proving $\pi_1(Z_0) = 1$.

We denote by Z_{ext} the complement of the interiors of mutually disjoint conical neighborhoods of the singular components, and we set $F = \text{Fix}(\sigma) \subset W$. Since F has codimension four in the connected manifold W , ordinary relative transversality shows that $W \setminus F$ is path connected. Its quotient is the regular locus of Z_0 . Radial deformation in each punctured cone fibre moves paths out of the deleted tubular interiors and onto their boundaries; hence Z_{ext} is path connected. We choose a basepoint in Z_{ext} and compatible basepaths to the boundary components. For one component B of F , which we identify with its image in Z_0 , we denote the boundary $\mathbb{RP}^3$ -bundle by $L \rightarrow B$, the cone bundle by $P_0 \rightarrow B$, and the truncated Eguchi–Hanson bundle by $P_1 \rightarrow B$; the latter is the bundle in (84). Their fibres are $\mathbb{RP}^3$, $C(\mathbb{RP}^3)$, and the disc bundle of $\mathcal{O}_{\mathbb{CP}^1}(-2)$, respectively. The last two fibres are connected and simply connected. The homotopy exact sequences therefore give isomorphisms

$$(p_i)_* : \pi_1(P_i) \xrightarrow{\cong} \pi_1(B), \quad i = 0, 1.$$

If $j_i : L \rightarrow P_i$ are the boundary inclusions and $p_L : L \rightarrow B$ is the projection, then $(p_i)_*(j_i)_* = (p_L)_*$. After identifying both $\pi_1(P_i)$ with $\pi_1(B)$ by $(p_i)_*$, the two boundary homomorphisms are therefore the same homomorphism $(p_L)_*$. This argument neither assumes that $\pi_1(\mathbb{RP}^3) \rightarrow \pi_1(L)$ is injective nor ignores monodromy. Van Kampen applies to open collar thickenings of Z_{ext} and P_i , whose intersection deformation retracts onto L , and gives the pushouts

$$\pi_1(Z_{\text{ext}} \cup_L P_i) \cong \pi_1(Z_{\text{ext}}) *_{\pi_1(L)} \pi_1(P_i), \quad i = 0, 1, \quad (131)$$

with identical structure maps after the preceding identifications. The fibrewise construction in Lemma 4.2 glues P_1 to Z_{ext} by the same fixed boundary identification used to glue P_0 . Therefore the two groups in (131) are isomorphic through the specified basepoint, basepath, and bundle-projection identifications.

A compact smooth fixed set has finitely many components. Enumerate them as $B_1, \dots, B_k$. For $0 \leq j \leq k$, let $Z^{(j)}$ be the space in which the cone bundles over $B_1, \dots, B_j$ have been replaced by their truncated Eguchi–Hanson bundles, while the cone bundles over the remaining components have not been replaced. Choose all basepaths in the common exterior. Applying the one-component pushout calculation to $Z^{(j-1)}$ and $Z^{(j)}$ gives an isomorphism $\pi_1(Z^{(j-1)}) \cong \pi_1(Z^{(j)})$ through those specified identifications. Induction on j therefore gives $\pi_1(\widehat{Z}) \cong \pi_1(Z_0) = 1$. $\square$

Theorem 6.2. *For every sufficiently small $t > 0$, the metric $g_t = g_{\widehat{\mathbb{F}}_t}$ has full Riemannian holonomy $\text{Hol}(\widehat{Z}, g_t) = \text{Spin}(7)$.*

Proof. We fix a sufficiently small t and set $g_t = g_{\widehat{\mathbb{F}}_t}$. A torsion-free $\text{Spin}(7)$ structure makes g_t Ricci-flat and determines a parallel unit positive spinor. Since $\widehat{Z}$ is compact, g_t is complete. Since $\widehat{Z}$ is simply connected, full and restricted holonomy agree. Moreover, $\pi_1(\widehat{Z}) = 0$ implies $H_1(\widehat{Z}; \mathbb{Z}) = 0$ by abelianization, and the universal coefficient theorem gives $H^1(\widehat{Z}; \mathbb{R}) = 0$. A parallel vector would dualize to a nonzero parallel one-form. Such a form is harmonic and represents a nonzero cohomology class, which is impossible. Hence the de Rham decomposition has no Euclidean factor.

The global de Rham decomposition theorem applies because $(\widehat{Z}, g_t)$ is complete and simply connected. Since there is no Euclidean factor, it gives a Riemannian product

$$(\widehat{Z}, g_t) = (X_1, g_1) \times \cdots \times (X_r, g_r)$$

of irreducible, complete, simply connected, nonflat factors. Each X_i is spin: restricting $w_2(T\widehat{Z}) = 0$ to X_i times basepoints in the other factors gives $w_2(TX_i) = 0$. We denote the restricted holonomy group of X_i by $H_i \subset SO(n_i)$. The factor spin structure determines its connected spin-holonomy lift $\widetilde{H}_i \subset \text{Spin}(n_i)$.

Under the standard homomorphism $\text{Spin}(n_1) \times \cdots \times \text{Spin}(n_r) \rightarrow \text{Spin}(8)$, the restriction of the complex positive spin module is a finite direct sum of external tensor products of complex factor spin modules. More explicitly, an odd-dimensional factor contributes its irreducible complex spin module, an even-dimensional factor contributes one of its two half-spin modules, and the parity constraint selects the summands; when two odd-dimensional factors are combined, the usual two parity summands occur. This is the standard branching rule for complex Clifford modules, and every projection onto one of the displayed tensor summands is equivariant for $\widetilde{H}_1 \times \cdots \times \widetilde{H}_r$. Therefore the parallel positive spinor has a nonzero invariant component in at least one tensor summand. If Δ_i denotes the factor spin or half-spin module occurring in that summand, then

$$(\Delta_1 \otimes \cdots \otimes \Delta_r)^{\widetilde{H}_1 \times \cdots \times \widetilde{H}_r} = \Delta_1^{\widetilde{H}_1} \otimes \cdots \otimes \Delta_r^{\widetilde{H}_r}.$$

The nonzero invariant component therefore implies $\Delta_i^{\widetilde{H}_i} \neq 0$ for every i . By the holonomy principle, every factor X_i therefore admits a nonzero parallel spinor.

The Berger–Simons–Wang classification [12] now applies to each factor. An irreducible nonsymmetric factor with a parallel spinor has holonomy $SU(m)$, $Sp(k)$, G_2 , or $\text{Spin}(7)$ in real dimension $2m$, $4k$, 7, or 8, respectively. An irreducible locally symmetric space is Einstein with nonzero Einstein constant unless it is Euclidean. Consequently an irreducible Ricci-flat symmetric factor is flat and has already been excluded. The remaining dimension possibilities are

factor dimension	nonflat parallel-spinor holonomy	required complementary dimension
8	$\text{Spin}(7)$, $SU(4)$, $Sp(2)$	0
7	G_2	1 (flat)
6	$SU(3)$	2 (flat)
4	$Sp(1)$	4.

The Wang list has no nonflat irreducible factor in dimension five, and Ricci-flat factors of dimensions one, two, or three are flat. Consequently a six- or seven-dimensional factor forces a flat two- or one-dimensional complement. Without a flat factor, the only possibilities are one eight-dimensional factor with holonomy $\text{Spin}(7)$, $SU(4)$, or $Sp(2)$, or two four-dimensional factors with holonomy $Sp(1) \times Sp(1)$. The de Rham product above is global, and full holonomy equals restricted holonomy because $\widehat{Z}$ is simply connected. Thus no factor-permuting or other disconnected extension of these connected holonomy groups occurs.

The projection of the compact product $\widehat{Z}$ onto each de Rham factor is surjective, so every factor is compact. Holonomy $SU(4)$ is Kähler, and holonomy $Sp(2)$ is hyperkähler. In the $Sp(1) \times Sp(1)$ case, de Rham gives a product of two compact hyperkähler four-manifolds, which is again Kähler. Every compact Kähler manifold is formal by [5], contradicting Theorem 5.5. Therefore $\text{Hol}(\widehat{Z}, g_t) = \text{Spin}(7)$. $\square$

Under the identification of Lemma 4.2, the map appearing in Theorem 1.1 is

$$\pi = q_0 \circ \rho : \widehat{Z} \longrightarrow \mathcal{O}.$$

It is the same map as the Joyce blowdown (58). Its properness was proved immediately after (58), and the local model is the identity over the regular stratum, so it is a diffeomorphism over the regular locus. Let ε_J be the upper bound supplied by Theorem 3.3, and let ε_H be the smallness bound in Theorem 6.2. Taking $\varepsilon = \min\{\varepsilon_J, \varepsilon_H\}$ makes the existence and full-holonomy conclusions hold for every $0 < t < \varepsilon$. Theorem 5.5 and Proposition 6.1 supply the remaining two assertions. This proves Theorem 1.1.

A Lean formalization of the algebraic deductions

The formalization encodes the algebraic implications used in the Massey, fundamental-group, and holonomy arguments. The geometric statements proved in the body of the paper enter through explicitly named hypotheses.

A.1 Affine cosets and the exceptional-coordinate obstruction

Let V be an abelian group, let $I \leq V$, and let

$$\mathcal{C}(y, I) = y + I = \{y + i : i \in I\}.$$

The definitions `AdditiveCore`, `AdditiveSubset`, and `affineCoset` encode these data. The theorem `zero_mem_affineCoset_iff` proves

$$0 \in \mathcal{C}(y, I) \iff y \in I. \tag{132}$$

Indeed, an equality $0 = y + i$ with $i \in I$ gives $y = -i \in I$, and the converse follows from $0 = y + (-y)$. Consequently, the theorem `full_affine_coset_exclusion` proves

$$y \notin I \implies 0 \notin \mathcal{C}(y, I).$$

If $y' \in y + I$ and $y \notin I$, then $y' \notin I$ as well. This implication is the theorem

$$\text{translated_representative_exclusion.}$$

For the exceptional projection in Section 5, write the coordinates in the order (v_1, v_2, v_3, v_7) . The projected indeterminacy is

$$J = \{(A, A, -B, B) : A, B \in \mathbb{R}\}, \tag{133}$$

whereas the projected representative is

$$y_{\text{MM}} = (-4, 4, 0, 0).$$

If $y_{\text{MM}} \in J$, its first two coordinates give $A = -4$ and $A = 4$. Thus $y_{\text{MM}} \notin J$. The theorem `unequal_coordinate_certificate` proves this implication for an arbitrary coefficient type from the inequality of the two coordinates, and `yMM_not_in_projected_indeterminacy` gives the corresponding decidable calculation over $\mathbb{Z}$. The theorem `exclusion_by_projection` then lifts the coordinate obstruction from J to the full indeterminacy subgroup.

A.2 Passage through the quotient and the resolution

The quotient step is expressed by groups V, Q , maps $\iota_u : V \rightarrow Q$ and $p_u : Q \rightarrow V$, and indeterminacy subgroups $I_V \leq V$ and $I_Q \leq Q$. The hypotheses are

$$p_u \iota_u = \text{id}_V, \quad p_u(I_Q) \subseteq I_V. \quad (134)$$

The theorem `quotient_indeterminacy_transfer` proves

$$y \notin I_V \implies \iota_u(y) \notin I_Q. \quad (135)$$

This is the formal counterpart of projection to the u -summand in Proposition 5.3.

For the resolution step, let $\rho^* : Q \rightarrow R$ and $\rho_! : R \rightarrow Q$ satisfy

$$\rho_! \rho^* = \text{id}_Q, \quad \rho_!(I_R) \subseteq I_Q. \quad (136)$$

The structure `GysinFormulaData` records separately the two terms in the triple-product indeterminacy and the corresponding projection formulas. The theorems `gysin_maps_full_indeterminacy` and `gysin_indeterminacy_transfer` prove

$$q \notin I_Q \implies \rho^* q \notin I_R. \quad (137)$$

The structure `MasseyPipelineInput` combines (132)–(137). Its conclusion is the theorem

$$\text{full_massey_coset_excludes_zero},$$

which states that zero is absent from the full Massey coset on the resolved space.

A.3 The dihedral and holonomy deductions

The infinite dihedral group is represented in normal form by

$$D_\infty = \{S^n : n \in \mathbb{Z}\} \sqcup \{S^n R : n \in \mathbb{Z}\}, \quad R^2 = 1, \quad RSR = S^{-1}.$$

If a normal subgroup $N \triangleleft D_\infty$ contains R and SR , then $S = (SR)R \in N$. Hence $S^n \in N$ and $S^n R \in N$ for every $n \in \mathbb{Z}$, so $N = D_\infty$. This is the theorem `DInf.normal_generation`; the structure `ArmstrongInput` combines it with the hypotheses of Armstrong's theorem and the invariance of the fundamental group under the fibrewise resolution.

The toric calculation is encoded using doubled support values. From $0 < \ell$ and $2\ell < L$, the theorem `toric_support_sign_certificate` proves

$$-(0 + 0 - 2\ell) = 2\ell > 0, \quad -(\ell + \ell - L) = L - 2\ell > 0.$$

These are the two forms of wall pairing needed in Section 3.2.

Finally, the formalized holonomy alternative is

$$\text{Spin}(7), \quad SU(4), \quad Sp(2), \quad Sp(1) \times Sp(1).$$

The three proper alternatives imply that the manifold is Kähler, and the DGMS theorem then implies formality. Under the hypothesis of nonformality, `holonomy_alternative_elimination` therefore selects `Spin(7)`.

A.4 Assembly

The structure `GeometricInput` contains the following propositions: the resolution is smooth and compact; its blowdown is proper and is a diffeomorphism over the regular locus; it carries a torsion-free $\text{Spin}(7)$ structure; the cohomological data satisfy the Massey pipeline; the covering action satisfies the hypotheses of Armstrong’s theorem; the fibrewise resolution preserves the fundamental group; the holonomy belongs to the four alternatives above; the three proper alternatives are Kähler; compact Kähler manifolds are formal; and formality forces zero to belong to every defined Massey product. These propositions are exactly the geometric, topological, and cohomological results established in the body or supplied by the cited theorems.

For every value D of `GeometricInput`, the theorem `verified_conclusion` produces the conjunction that the resolved space is smooth, compact, and simply connected; the blowdown is proper and is a diffeomorphism over the regular locus; the torsion-free structure has full holonomy $\text{Spin}(7)$; and the full Massey coset excludes zero, so the resolved space is nonformal. This is the formalized logical assembly of Theorem 1.1.

References

- [1] M. A. Armstrong, *The fundamental group of the orbit space of a discontinuous group*, Proc. Cambridge Philos. Soc. **64** (1968), 299–301.
- [2] A. K. Bousfield and V. K. A. M. Gugenheim, *On PL de Rham theory and rational homotopy type*, Mem. Amer. Math. Soc. **8** (1976), no. 179.
- [3] G. E. Bredon, *Sheaf Theory*, second ed., Graduate Texts in Mathematics 170, Springer, New York, 1997.
- [4] G. E. Bredon, *Introduction to Compact Transformation Groups*, Pure and Applied Mathematics 46, Academic Press, New York–London, 1972.
- [5] P. Deligne, P. Griffiths, J. Morgan, and D. Sullivan, *Real homotopy theory of Kähler manifolds*, Invent. Math. **29** (1975), 245–274.
- [6] J. L. Dupont, *Curvature and Characteristic Classes*, Lecture Notes in Mathematics 640, Springer, Berlin, 1978.
- [7] W. Fulton, *Introduction to Toric Varieties*, Annals of Mathematics Studies 131, Princeton University Press, Princeton, 1993.
- [8] S. Illman, *The equivariant triangulation theorem for actions of compact Lie groups*, Math. Ann. **262** (1983), 487–501.
- [9] M. W. Hirsch, *Differential Topology*, Graduate Texts in Mathematics 33, Springer, New York, 1976.
- [10] D. D. Joyce, *Compact Manifolds with Special Holonomy*, Oxford Mathematical Monographs, Oxford University Press, 2000.
- [11] L. Martín-Merchán, *Compact holonomy G_2 manifolds need not be formal*, Geom. Topol. **30** (2026), 1109–1127; doi:10.2140/gt.2026.30.1109.
- [12] M. Y. Wang, *Parallel spinors and parallel forms*, Ann. Global Anal. Geom. **7** (1989), 59–68; doi:10.1007/BF00137402.